Physica bk6

May 8, 2026

1 On GlossySnake

A LaTeX Proof-of-Concept of the GlossySnake project. By the nature of the machine-glossing, the inaccurate and malformatted glossing is expected.

2 Physica bk6

PHYSICA Bk.VI

PHYSICS Book VI

Κεφάλαιο 1

Chapter 1

Εἰ δ' ἐστὶ συνεχὲς καὶ ἁπτόμενον καὶ ἐφεξῆς, ὡς διώρισται

If but it is continuous and touching and subsequently, as has been defined

πρότερον, συνεχῆ μὲν ὧν τὰ ἔσχατα ἕν, ἁπτόμενα

previously, continuous on one hand of which the extremes are one, touching

δ' ὧν ἅμα, ἐφεξῆς δ' ὧν μηδὲν μεταξὺ συγγενές,

but of which together, successively but of which nothing in between is related,

ἀδύνατον ἐξ ἀδιαιρέτων εἶναί τι συνεχές, οἷον γραμμὴν

impossible from indivisible to be something continuous, such as a line

ἐκ στιγμῶν, εἴπερ ἡ γραμμὴ μὲν συνεχές, ἡ στιγμὴ δὲ

made of points, if ever the line indeed is continuous, the point but

ἀδιαίρετον. οὔτε γὰρ ἓν τὰ ἔσχατα τῶν στιγμῶν (οὐ γάρ ἐστι

is indivisible. Neither for one of the extremes of the points (not for it is

τὸ μὲν ἔσχατον τὸ δ' ἄλλο τι μόριον τοῦ ἀδιαιρέτου),

the on one hand extreme the but another something portion of the indivisible),

οὔθ' ἅμα τὰ ἔσχατα (οὐ γάρ ἐστιν ἔσχατον τοῦ ἀμεροῦς

nor at the same time the extremes (not for it is an extreme of the unmeasurable

οὐδέν· ἕτερον γὰρ τὸ ἔσχατον καὶ οὗ ἔσχατον).

nothing. another for the ultimate and of which ultimate).

ἔτι δ' ἀνάγκη ἤτοι συνεχεῖς εἶναι τὰς στιγμὰς ἢ ἁπτομένας

still but necessity either continuous to be the moments or touching

ἀλλήλων, ἐξ ὧν ἐστι τὸ συνεχές· ὁ δ' αὐτὸς λόγος καὶ ἐπὶ

each other, from which is the continuous· the but same reason and upon

πάντων τῶν (231b.) ἀδιαιρέτων. συνεχεῖς μὲν δὴ οὐκ ἂν

all of the (231b.) indivisible. continuous indeed certainly not would

εἶεν διὰ τὸν εἰρημένον λόγον· ἅπτεται δ' ἅπαν ἢ ὅλον

be because of the stated reason· it touches but all or whole

ὅλου ἢ μέρος μέρους ἢ ὅλου μέρος. ἐπεὶ δ' ἀμερὲς

of the whole or part of a part or of the whole part. since but indivisible

τὸ ἀδιαίρετον, ἀνάγκη ὅλον ὅλου ἅπτεσθαι. ὅλον δ'

the undivisible, necessity to touch the whole of the whole to touch. whole but

ὅλου ἁπτόμενον οὐκ ἔσται συνεχές. τὸ γὰρ συνεχὲς ἔχει τὸ

of the whole touching not will be continuous. the for continuous has the

μὲν ἄλλο τὸ δ' ἄλλο μέρος, καὶ διαιρεῖται εἰς οὕτως ἕτερα καὶ

indeed other the but another part, and is divided into thus other and

τόπῳ κεχωρισμένα.

place separated.

ἀλλὰ μὴν οὐδὲ ἐφεξῆς ἔσται στιγμὴ στιγμῇ ἢ τὸ

but on the other hand nor subsequently will be moment in the moment or the

νῦν τῷ νῦν, ὥστ' ἐκ τούτων εἶναι τὸ μῆκος ἢ τὸν χρόνον·

now to the now, so that from these to be the length or the time.

ἐφεξῆς μὲν γάρ ἐστιν ὧν μηθέν ἐστι μεταξὺ συγγενές,

subsequently indeed for there is of which nothing is between related,

στιγμῶν δ' αἰεὶ [τὸ] μεταξὺ γραμμὴ καὶ τῶν νῦν χρόνος. ἔτι

moments but always [the] between line and of the now time. also

διαιροῖτ' ἂν εἰς ἀδιαίρετα, εἴπερ ἐξ ὧν ἐστιν ἑκάτερον, εἰς

will divide if into indivisible, if indeed of of which there is each, into

ταῦτα διαιρεῖται· ἀλλ' οὐθὲν ἦν τῶν συνεχῶν εἰς ἀμερῆ διαιρετόν.

these is divisible; but nothing was of the continuous into parts divisible.

ἄλλο δὲ γένος οὐχ οἷόν τ' εἶναι μεταξὺ [τῶν στιγμῶν καὶ τῶν νῦν

another but kind not able to be between [the moments and the now

οὐθέν]. ἢ γὰρ [ἔσται, δῆλον ὡς ἤτοι] ἀδιαίρετον ἔσται ἢ διαιρετόν,

nothing]. or for [will be, clear that either] undivided will be or divisible,

καὶ εἰ διαιρετόν, ἢ εἰς ἀδιαίρετα ἢ εἰς ἀεὶ διαιρετά· τοῦτο δὲ

and if divisible, or into undivided or into always divisible. this but

συνεχές.

continuous.

φανερὸν δὲ καὶ ὅτι πᾶν συνεχὲς διαιρετὸν εἰς αἰεὶ διαιρετά· εἰ

clear but and that all continuous divisible into always divisible. if

γὰρ εἰς ἀδιαίρετα, ἔσται ἀδιαίρετον ἀδιαιρέτου ἁπτόμενον· ἓν γὰρ

for into undivided, will be undivided of undivided touching. one for

τὸ ἔσχατον καὶ ἅπτεται τῶν συνεχῶν.

the last and touches of the continuous.

τοῦ δ' αὐτοῦ λόγου μέγεθος καὶ χρόνον καὶ κίνησιν ἐξ ἀδιαιρέτων

of the and its word magnitude and time and motion from indivisibles

συγκεῖσθαι, καὶ διαιρεῖσθαι εἰς ἀδιαίρετα, ἢ μηθέν. δῆλον
to be composed, and to be divided into indivisible entities, or nothing. It is clear
δ' ἐκ τῶνδε. εἰ γὰρ τὸ μέγεθος ἐξ ἀδιαιρέτων σύγκειται, καὶ
and from these. If for the magnitude from indivisibles is composed, and
ἡ κίνησις ἡ τούτου ἐξ ἴσων κινήσεων ἔσται ἀδιαιρέτων, οἷον
the motion the of this from equal movements will be indivisibles, like
εἰ τὸ ΑΒΓ ἐκ τῶν Α Β Γ ἐστὶν ἀδιαιρέτων, ἡ κίνησις ἐφ'
if the ABC from the A B C is indivisible, the motion on which
ἧς ΔΕΖ, ἣν ἐκινήθη τὸ Ω ἐπὶ τῆς ΑΒΓ, ἕκαστον τὸ μέρος
it DEZ, which moved the Omega on the ABC, each the part
ἔχει ἀδιαίρετον. εἰ δὴ παρούσης κινήσεως ἀνάγκη κινεῖσθαί
has indivisible. If indeed of the present motion necessity to be moved
τι, καὶ εἰ κινεῖταί τι, παρεῖναι κίνησιν, καὶ τὸ
by something, and if it is moved by something, to be present motion, and the
κινεῖσθαι ἔσται ἐξ ἀδιαιρέτων. τὸ μὲν δὴ Α ἐκινήθη τὸ
to be moved will be from indivisibles. the on one hand indeed A was moved the
Ω τὴν τὸ Δ κινούμενον κίνησιν, τὸ δὲ Β τὴν τὸ Ε, καὶ τὸ Γ
Ω the the D being moved motion, the but B the the E, and the G
ὡσαύτως τὴν τὸ Ζ. εἰ δὴ ἀνάγκη τὸ κινούμενον ποθέν ποι
likewise the the Z. if indeed necessity the being moved from somewhere to be
μὴ ἅμα κινεῖσθαι καὶ κεκινῆσθαι οὗ ἐκινεῖτο ὅτε
not at the same time to be moved and to have been moved where it was moving when
ἐκινεῖτο (οἷον εἰ Θήβαζέ τι βαδίζει, ἀδύνατον ἅμα
it was moving (like if Thebes something is walking, impossible at the same time
βαδίζειν Θήβαζε καὶ βεβαδικέναι [232a] Θήβαζε), τὴν δὲ τὸ Α τὴν
to walk Thebes and to have walked [232a] Thebes), the but the A the
ἀμερῆ ἐκινεῖτο τὸ Ω, ᾗ ἡ τὸ Δ κίνησις παρῆν· ὥστ'
indivisible it would be moved the Ω, in which the the D motion was present; so that

εἰ μὲν ὕστερον διεληλύθει ἢ διήει, διαιρετὴ ἂν

if on one hand later it has been separated or it goes through, separable if

εἴη (ὅτε γὰρ διήει, οὔτε ἠρέμει οὔτε διεληλύθει, ἀλλὰ

it were (when for it goes through, neither it rests nor has been separated, but

μεταξὺ ἦν), εἰ δ' ἅμα διέρχεται καὶ διελήλυθε,

in between it was), if and at the same time it passes through and has been separated,

τὸ βαδίζον, ὅτε βαδίζει, βεβαδικὸς ἐκεῖ ἔσται καὶ κεκινημένον

the walking, when it walks, one that has walked there will be and moved

οὗ κινεῖται. εἰ δὲ τὴν μὲν ὅλην τὴν ΑΒΓ κινεῖταί τι,

where it is moved. if but the one whole the ABC is being moved something,

καὶ ἡ κίνησις ἣν κινεῖται τὰ Δ Ε Ζ ἐστι, τὴν δ' ἀμερῆ

and the motion which is being moved the D E Z is, the and unseparated

τὴν Α οὐθὲν κινεῖται ἀλλὰ κεκίνηται, εἴη ἂν ἡ κίνησις οὐκ

the A nothing is moved but has been moved, it would be if the motion not

ἐκ κινήσεων ἀλλ' ἐκ κινημάτων καὶ τῷ κεκινῆσθαί τι μὴ

from movements but from motions and to the having been moved some not

κινούμενον· τὴν γὰρ Α διελήλυθεν οὐ διεξιόν. ὥστε

being moved. the for A has been separated not it must be carried out. so that

ἔσται τι βεβαδικέναι μη δέποτε βαδίζον· ταύτην γὰρ

it will be something to have been established not ever walking. this for

βεβάδικεν οὐ βαδίζον ταύτην. εἰ οὖν ἀνάγκη ἢ ἠρεμεῖν ἢ

has established not walking this. if therefore necessity or to be at rest or

κινεῖσθαι πᾶν, ἠρεμεῖ καθ' ἕκαστον τῶν Α Β Γ,

to be in motion everything, let it be at rest according to each of the A B C,

ὥστ' ἔσται τι συνεχῶς ἠρεμοῦν ἅμα καὶ κινούμενον.

so that it will be something continuously being at rest together and moving.

τὴν γὰρ ΑΒΓ ὅλην ἐκινεῖτο καὶ ἠρέμει ὁτιοῦν μέρος,

the for ABC whole it would move and it would be at rest any part,

ὥστε καὶ πᾶσαν. καὶ εἰ μὲν τὰ ἀδιαίρετα τῆς ΔΕΖ κινήσεις,

so that and all. and if indeed the indivisible of the DEZ movements,

κινήσεως παρούσης ἐνδέχοιτ' ἂν μὴ κινεῖσθαι ἀλλ' ἠρεμεῖν·

of motion being present it might happen that not to be in motion but to be at rest.

εἰ δὲ μὴ κινήσεις, τὴν κίνησιν μὴ ἐκ κινήσεων εἶναι. ὁμοίως δ'

if but not movements, the motion not from movements to be. similarly and

ἀνάγκη τῷ μήκει καὶ τῇ κινήσει ἀδιαίρετον εἶναι τὸν χρόνον, καὶ

necessity for the length and the motion indivisible to be the time, and

συγκεῖσθαι ἐκ τῶν νῦν ὄντων ἀδιαιρέτων· εἰ γὰρ πᾶσα διαιρετός,

to consist of the now existing indivisible. if for every divisible,

ἐν τῷ ἐλάττονι δὲ τὸ ἰσοταχὲς δίεισιν ἔλαττον, διαιρετὸς

in the lesser but the equal passes lesser, it will be divisible

ἔσται καὶ ὁ χρόνος. εἰ δ' ὁ χρόνος διαιρετὸς ἐν ᾧ φέρεταί

will be and the time. if then the time is divisible in which is carried

τι τὴν Α, καὶ ἡ τὸ Α ἔσται διαιρετή.

something the A, and the the A will be divisible.

Κεφάλαιο 2

Chapter 2

Ἐπεὶ δὲ πᾶν μέγεθος εἰς μεγέθη διαιρετόν (δέδεικται γὰρ ὅτι

Since and every magnitude into magnitudes divisible (it has been shown for that

ἀδύνατον ἐξ ἀτόμων εἶναί τι συνεχές, μέγεθος δ' ἐστὶν

impossible from atoms to be something continuous, magnitude but it is

ἅπαν συνεχές), ἀνάγκη τὸ θᾶττον ἐν τῷ ἴσῳ χρόνῳ μεῖζον καὶ

all continuous), necessity the faster in the equal time greater and

ἐν τῷ ἐλάττονι ἴσον καὶ ἐν τῷ ἐλάττονι πλεῖον κινεῖσθαι, καθάπερ

in the lesser equal and in the lesser more to be moved, as

ὁρίζονταί τινες τὸ θᾶττον. ἔστω γὰρ τὸ ἐφ' ᾧ Α τοῦ ἐφ'

are defined some the faster. Let it be for the on which that A of on which

ᾧ Β θᾶττον. ἐπεὶ τοίνυν θᾶττόν ἐστιν τὸ πρότερον μεταβάλλον,

that B is faster. Since therefore faster it is the earlier changing,

ἐν ᾧ χρόνῳ τὸ Α μεταβέβληκεν ἀπὸ τοῦ Γ εἰς τὸ Δ, οἷον

in which time the A has changed from the G into the D, such as

τῷ ΖΗ, ἐν τούτῳ τὸ Β οὔπω ἔσται πρὸς τῷ Δ, ἀλλ' ἀπολείψει,

the ZH, in this the B not yet will be towards the D but will leave

ὥστε ἐν τῷ ἴσῳ χρόνῳ πλεῖον δίεισιν τὸ θᾶττον. ἀλλὰ μὴν καὶ

so that in the equal time more will pass the faster but indeed and

ἐν τῷ ἐλάττονι πλεῖον· ἐν ᾧ γὰρ τὸ Α γεγένηται πρὸς

in the lesser more in which for the A has been generated towards

τῷ Δ, τὸ Β ἔστω πρὸς τῷ Ε τὸ βραδύτερον ὄν. οὐκοῦν ἐπεὶ

the D the B let it be towards the E the slower being therefore since

[232b] τὸ Α πρὸς τῷ Δ γεγένηται ἐν ἅπαντι τῷ ΖΗ χρόνῳ,

[232b] the A towards the D has been generated in all the ZH time

πρὸς τῷ Θ ἔσται ἐν ἐλάττονι τούτου· καὶ ἔστω ἐν τῷ ΖΚ.

towards the Θ will be in lesser of this and let it be in the ZK

τὸ μὲν οὖν ΓΘ, ὃ διελήλυθε τὸ Α, μεῖζόν ἐστι

the on the one hand therefore GΘ which has been separated the A greater is

τοῦ ΓΕ, ὁ δὲ χρόνος ὁ ΖΚ ἐλάττων τοῦ παντὸς τοῦ ΖΗ,

of the GE the but time the ZK lesser of the whole of the ZH,

ὥστε ἐν ἐλάττονι μεῖζον δίεισιν. φανερὸν δὲ ἐκ τούτων καὶ ὅτι

so that in lesser greater it passes. evident but from these and that

τὸ θᾶττον ἐν ἐλάττονι χρόνῳ δίεισιν τὸ ἴσον. ἐπεὶ γὰρ τὴν μείζω

the swifter in lesser time it passes the same. since for the greater

ἐν ἐλάττονι διέρχεται τοῦ βραδυτέρου, αὐτὸ δὲ καθ' αὑτὸ
in lesser it passes of the slower, itself but according to itself
λαμβανόμενον ἐν πλείονι χρόνῳ τὴν μείζω τῆς ἐλάττονος, οἷον
taken in larger time the greater of the lesser, such as
τὴν ΛΜ τῆς ΛΞ, πλείων ἂν εἴη ὁ χρόνος ὁ ΠΡ, ἐν ᾧ τὴν
the LM of the LX, greater would be the time the PR, in which the
ΛΜ διέρχεται, ἢ ὁ ΠΣ, ἐν ᾧ τὴν ΛΞ. ὥστε εἰ ὁ ΠΡ χρόνος
LM it passes, or the PS, in which the LX. so that if the PR time
ἐλάττων ἐστὶν τοῦ Χ, ἐν ᾧ τὸ βραδύτερον διέρχεται τὴν ΛΞ,
lesser is of the X, in which the slower passes the ΛΞ,
καὶ ὁ ΠΣ ἐλάττων ἔσται τοῦ ἐφ' ᾧ Χ· τοῦ γὰρ ΠΡ ἐλάττων,
and the ΠΣ lesser will be of the on which X· of the for ΠΡ lesser,
τὸ δὲ τοῦ ἐλάττονος ἔλαττον καὶ αὐτὸ ἔλαττον. ὥστε ἐν ἐλάττονι
the but of the lesser less and itself less. so that in lesser
κινήσεται τὸ ἴσον. ἔτι δ' εἰ πᾶν ἀνάγκη ἢ ἐν ἴσῳ ἢ ἐν ἐλάττονι ἢ
will move the equal. yet but if all necessity or in equal or in lesser or
ἐν πλείονι κινεῖ σθαι, καὶ τὸ μὲν ἐν πλείονι βραδύτερον, τὸ
in greater moves to be, and the on the one hand in greater slower, the
δ' ἐν ἴσῳ ἰσοταχές, τὸ δὲ θᾶττον οὔτε ἰσοταχὲς οὔτε βραδύτερον,
but in equal equidistant, the but faster neither equidistant nor slower,
οὔτ' ἂν ἐν ἴσῳ οὔτ' ἐν πλείονι κινοῖτο τὸ θᾶττον. λείπεται
nor if in equal nor in greater it would move the faster. is left
οὖν ἐν ἐλάττονι, ὥστ' ἀνάγκη καὶ τὸ ἴσον μέγεθος ἐν ἐλάττονι
therefore in less so that necessity and the equal magnitude in less
χρόνῳ διιέναι τὸ θᾶττον. ἐπεὶ δὲ πᾶσα μὲν κίνησις ἐν χρόνῳ
time to pass the faster. since but every indeed motion in time
καὶ ἐν ἅπαντι χρόνῳ δυνατὸν κινηθῆναι, πᾶν δὲ τὸ κινούμενον
and in all time possible to be moved, every but the moving

ἐνδέχεται καὶ θᾶττον κινεῖσθαι καὶ βραδύτερον, ἐν ἅπαντι χρόνῳ

is likely and faster to be moved and slower, in all time

ἔσται τὸ θᾶττον κινεῖσθαι καὶ βραδύτερον. τούτων δ' ὄντων

will be the faster to be moved and slower. of these and being

ἀνάγκη καὶ τὸν χρόνον συνεχῆ εἶναι. λέγω δὲ συνεχὲς τὸ διαιρετὸν

necessity and the time continuous to be. I say but continuous the divisible

εἰς αἰεὶ διαιρετά· τούτου γὰρ ὑποκειμένου τοῦ συνε χοῦς,

into always divisible; of this for underlying the continuous substance,

ἀνάγκη συνεχῆ εἶναι τὸν χρόνον. ἐπεὶ γὰρ δέδεικται ὅτι τὸ

necessity continuous to be the time. since for it has been shown that the

θᾶττον ἐν ἐλάττονι χρόνῳ δίεισιν τὸ ἴσον, ἔστω τὸ μὲν

faster in lesser time passes the equal, let it be the on the one hand

ἐφ' ᾧ Α θᾶττον, τὸ δ' ἐφ' ᾧ Β βραδύτερον, καὶ

in regard to which A faster, the but in regard to which B slower, and

κεκινήσθω τὸ βραδύτερον τὸ ἐφ' ᾧ ΓΔ μέγεθος ἐν τῷ

let it move the slower the in regard to which GD magnitude in the

ΖΗ χρόνῳ. δῆλον τοίνυν ὅτι τὸ θᾶττον ἐν ἐλάττονι τούτου

ZH time. it is clear therefore that the faster in lesser of this

κινήσεται τὸ αὐτὸ μέγεθος· καὶ κεκινήσθω ἐν τῷ ΖΘ. πάλιν δ'

will move the same magnitude; and let it move in the ZΘ. again but

ἐπεὶ τὸ θᾶττον ἐν τῷ ΖΘ διελήλυθεν τὴν ὅλην τὴν ΓΔ, τὸ

since the faster in the ZΘ has passed through the whole the GD, the

βραδύτερον ἐν τῷ αὐτῷ χρόνῳ τὴν ἐλάττω δίεισιν· ἔστω οὖν

slower in the same time the lesser passes; let it be therefore

ἐφ' [233a] ἧς ΓΚ. ἐπεὶ δὲ τὸ βραδύτερον τὸ Β ἐν τῷ ΖΘ

in regard to [233a] of which GK. since but the slower the B in the ZΘ

χρόνῳ τὴν ΓΚ διελήλυθεν, τὸ θᾶττον ἐν ἐλάττονι δίεισιν, ὥστε

time the GK has passed, the faster in lesser goes, so that

πάλιν διαιρεθήσεται ὁ ΖΘ χρόνος. τούτου δὲ διαιρουμένου καὶ τὸ

again will be divided the ΖΘ time. of this but being divided and the

ΓΚ μέγεθος διαιρεθήσεται κατὰ τὸν αὐτὸν λόγον. εἰ δὲ τὸ

ΓΚ magnitude will be divided according to the same reason. if but the

μέγεθος, καὶ ὁ χρόνος. καὶ ἀεὶ τοῦτ' ἔσται μεταλαμβάνουσιν ἀπὸ

magnitude, and the time. and always this will be will participate from

τοῦ θάττονος τὸ βραδύτερον καὶ ἀπὸ τοῦ βραδυτέρου τὸ θᾶττον,

the faster the slower and from the slower the faster,

καὶ τῷ ἀποδεδειγμένῳ χρωμένοις· διαιρήσει γὰρ τὸ μὲν θᾶττον

and the proven to those using· will divide for the on one hand faster

τὸν χρόνον, τὸ δὲ βραδύτερον τὸ μῆκος. εἰ οὖν αἰεὶ

the time, the on the other hand slower the length. if then always

μὲν ἀντιστρέφειν ἀληθές, ἀντιστρεφομένου δὲ αἰεὶ γίγνεται

on one hand to invert true, of being turned but always becomes

διαίρεσις, φανερὸν ὅτι πᾶς χρόνος ἔσται συνεχής.

division, evident that every time will be continuous.

ἅμα δὲ δῆλον καὶ ὅτι μέγεθος ἅπαν ἐστὶ συνεχές· τὰς

at the same time and evident and that magnitude all is continuous. the

αὐτὰς γὰρ καὶ τὰς ἴσας διαιρέσεις ὁ χρόνος διαιρεῖται καὶ τὸ

same for and the equal divisions the time is divided and the

μέγεθος. ἔτι δὲ καὶ ἐκ τῶν εἰωθότων λόγων λέγεσθαι φανερὸν ὡς

magnitude. also but and from the accustomed words to be said evident that

εἴπερ ὁ χρόνος ἐστὶ συνεχής, ὅτι καὶ τὸ μέγεθος, εἴπερ ἐν τῷ

if indeed the time is continuous, that and the magnitude, if indeed in the

ἡμίσει χρόνῳ ἥμισυ διέρχεται καὶ ἁπλῶς ἐν τῷ ἐλάττονι ἔλαττον·

half time half passes and simply in the lesser lesser.

αἱ γὰρ αὐταὶ διαιρέσεις ἔσονται τοῦ χρόνου καὶ τοῦ μεγέθους.

the for same divisions will be of the time and of the magnitude.

καὶ εἰ ὁποτερονοῦν ἄπειρον, καὶ θάτερον, καὶ ὡς θάτερον, καὶ

and if either infinite, and the other, and as the other, and

θάτερον, οἷον εἰ μὲν τοῖς ἐσχάτοις ἄπειρος ὁ χρόνος, καὶ τὸ

the other, such as if indeed the last infinite the time, and the

μῆκος τοῖς ἐσχάτοις, εἰ δὲ τῇ διαιρέσει, τῇ διαιρέσει καὶ τὸ μῆκος,

length to the last, if but the division, the division and the length,

εἰ δὲ ἀμφοῖν, ἀμφοῖν καὶ τὸ μέγεθος.

if but both, both and the size.

διὸ καὶ ὁ Ζήνωνος λόγος ψεῦδος λαμβάνει τὸ μὴ ἐνδέχεσθαι

therefore and the of Zeno argument is false takes the not to be possible

τὰ ἄπειρα διελθεῖν ἢ ἅψασθαι τῶν ἀπείρων καθ' ἕκαστον

the infinite to pass through or to touch of the infinity according to each

ἐν πεπερασμένῳ χρόνῳ. διχῶς γὰρ λέγεται καὶ τὸ μῆκος καὶ

in finite time. in two ways for it is said and the length and

ὁ χρόνος ἄπειρον, καὶ ὅλως πᾶν τὸ συνεχές, ἤτοι κατὰ

the time is infinite, and altogether all the continuous, or according to

διαίρεσιν ἢ τοῖς ἐσχάτοις. τῶν μὲν οὖν κατὰ τὸ

division or to the extremes. of the on the one hand therefore according to the

ποσὸν ἀπείρων οὐκ ἐνδέχεται ἅψασθαι ἐν πεπερασμένῳ χρόνῳ,

quantity is infinite not possible to touch in finite time,

τῶν δὲ κατὰ διαίρεσιν ἐνδέχεται· καὶ γὰρ αὐτὸς ὁ χρόνος

of the but according to division it is possible; for indeed the self the time

οὕτως ἄπειρος. ὥστε ἐν τῷ ἀπείρῳ καὶ οὐκ ἐν τῷ πεπερασμένῳ

in such a way is infinite. so that in the infinite and not in the finite

συμβαίνει διιέναι τὸ ἄπειρον, καὶ ἅπτεσθαι τῶν ἀπείρων τοῖς

it happens to pass through the infinite, and to touch of the infinite to the

ἀπείροις, οὐ τοῖς πεπερασμένοις. οὔτε δὴ τὸ ἄπειρον οἷόν τε ἐν

infinities, not to the finite. nor indeed the infinite such can in

πεπερασμένῳ χρόνῳ διελθεῖν, οὔτ' ἐν ἀπείρῳ τὸ πεπερασμένον·

finite time to pass, nor in infinite the finite;

ἀλλ' ἐάν τε ὁ χρόνος ἄπειρος ᾖ, καὶ τὸ μέγεθος ἔσται ἄπειρον,

but if it the time is infinite , and the magnitude will be infinite,

ἐάν τε τὸ μέγεθος, καὶ ὁ χρόνος. ἔστω γὰρ πεπερασμένον

if it the magnitude, and the time. let it be for a finite

μέγεθος ἐφ' οὗ AB, χρόνος δὲ ἄπειρος ἐφ' ᾧ Γ· εἰλήφθω

magnitude on which it AB, time but infinite on which it G. let it be taken

δέ τι τοῦ [233b] χρόνου πεπερασμένον, ἐφ' ᾧ ΓΔ.

also something of the [233b] time finite, on which it GD.

ἐν τούτῳ οὖν δίεισί τι τοῦ μεγέθους, καὶ ἔστω

in this therefore it divides something of the magnitude, and let it be

διεληλυθὸς ἐφ' ᾧ BE. τοῦτο δὲ ἢ καταμετρήσει τὸ ἐφ' ᾧ

divided on which it BE. this but or will measure the on which

AB, ἢ ἐλλείψει, ἢ ὑπερβαλεῖ· διαφέρει γὰρ οὐθέν· εἰ γὰρ ἀεὶ τὸ

AB, or in absence, or will exceed. differs for nothing; if for always the

ἴσον τῷ BE μέγεθος ἐν ἴσῳ χρόνῳ δίεισιν, τοῦτο δὲ καταμετρεῖ τὸ

equal to BE magnitude in equal time passes, this but measures the

ὅλον, πεπερασμέ νος ἔσται ὁ πᾶς χρόνος ἐν ᾧ διῆλθεν· εἰς

whole, finite quantity will be the all time in which it passed; into

ἴσα γὰρ διαιρεθήσεται καὶ τὸ μέγεθος. ἔτι δ' εἰ μὴ πᾶν μέγεθος

equals for will be divided and the magnitude. still but if not every magnitude

ἐν ἀπείρῳ χρόνῳ δίεισιν, ἀλλ' ἐνδέχεταί τι καὶ ἐν πεπερασμένῳ

in infinite time passes, but it may something and in finite

διελθεῖν, οἷον τὸ BE, τοῦτο δὲ καταμετρήσει τὸ πᾶν, καὶ τὸ ἴσον

to pass, such as the BE, this but will measure the whole, and the equal

ἐν ἴσῳ δίεισιν, ὥστε πεπερασμένος ἔσται καὶ ὁ χρόνος. ὅτι δ'

in equal passes, so that finite will be and the time. that but

οὐκ ἐν ἀπείρῳ δίεισιν τὸ BE, φανερόν, εἰ ληφθείη ἐπὶ θάτερα

not in infinite passes the BE, obvious, if it were to be taken on the other

πεπερασμένος ὁ χρόνος· εἰ γὰρ ἐν ἐλάττονι τὸ μέρος δίεισιν,

finite the time; if indeed in lesser the part passes,

τοῦτο ἀνάγκη πεπεράνθαι, θατέρου γε πέρατος ὑπάρχοντος.

this necessity to be finite, of the other indeed limit existing.

ἡ αὐτὴ δὲ ἀπόδειξις καὶ εἰ τὸ μὲν μῆκος ἄπειρον ὁ δὲ

the same but proof and if the on one hand length is infinite the but

χρόνος πεπερασμένος. φανερὸν οὖν ἐκ τῶν εἰρημένων ὡς

time is finite. evident therefore from the aforementioned that

οὔτε γραμμὴ οὔτε ἐπίπεδον οὔτε ὅλως τῶν συνεχῶν οὐθὲν ἔσται

neither line nor plane nor at all of the continuous nothing will be

ἄτομον, οὐ μόνον διὰ τὸ νῦν λεχθέν, ἀλλὰ καὶ ὅτι συμβήσεται

atom, not only by the now said, but also and that it will come to be

διαιρεῖσθαι τὸ ἄτομον. ἐπεὶ γὰρ ἐν ἅπαντι χρόνῳ τὸ θᾶττον καὶ

to be divided the atom. since indeed in all time the faster and

βραδύτερον ἔστι, τὸ δὲ θᾶττον πλεῖον διέρχεται ἐν τῷ ἴσῳ χρόνῳ,

slower is, the but faster more passes in the equal time,

ἐνδέχεται δὲ καὶ διπλάσιον καὶ ἡμιόλιον διιέναι μῆκος (εἴη

it may but also double and half to pass length (it would

γὰρ ἂν οὗτος ὁ λόγος τοῦ τάχους), ἐνηνέχθω οὖν τὸ

for would this the reason of the speed), let it be carried therefore the

θᾶττον ἡμιόλιον ἐν τῷ αὐτῷ χρόνῳ, καὶ διῃρήσθω τὰ μεγέθη

faster half in the same time, and let it be divided the magnitudes

τὸ μὲν τοῦ θάττονος εἰς τρία ἄτομα, ἐφ' ὧν ΑΒ ΒΓ ΓΔ,

the on one hand of the faster into three units, about which AB BC CD,

τὸ δὲ τοῦ βραδυτέρου εἰς δύο, ἐφ' ὧν ΕΖ ΖΗ. οὐκοῦν καὶ ὁ

the but of the slower into two, about which EF GH. Therefore also the

χρόνος διαιρεθήσεται εἰς τρία ἄτομα· τὸ γὰρ ἴσον ἐν τῷ ἴσῳ

time will be divided into three units. the for equal in to the same

χρόνῳ δίεισιν. διῃρήσθω οὖν ὁ χρόνος εἰς τὰ ΚΛ ΛΜ

time it divides. let it be divided therefore the time into the KL LM

ΜΝ. πάλιν δ' ἐπεὶ τὸ βραδύτερον ἐνήνεκται τὴν ΕΖΗ, καὶ ὁ

MN. again but since the slower is carried the EZH, and the

χρόνος τμηθήσεται δίχα. διαιρεθήσεται ἄρα τὸ ἄτομον, καὶ τὸ

time will be divided apart. will be separated therefore the atom, and the

ἀμερὲς οὐκ ἐν ἀτόμῳ δίει σιν ἀλλ' ἐν πλείονι. φανερὸν οὖν

indivisible not in atom divides it but in plurality. evident therefore

ὅτι οὐδέν ἐστι τῶν συνεχῶν ἀμερές.

that nothing is of the continuous indivisible.

Κεφάλαιο 3

Chapter 3

Ἀνάγκη δὲ καὶ τὸ νῦν τὸ μὴ καθ' ἕτερον ἀλλὰ καθ'

Necessity but and the now the not according to another but according to

αὐτὸ καὶ πρῶτον λεγόμενον ἀδιαίρετον εἶναι, καὶ ἐν ἅπαντι τὸ

itself and first said indivisible to be, and in all the

τοιοῦτο χρόνῳ ἐνυπάρχειν. ἔστιν γὰρ ἔσχατόν τι τοῦ [234a]

such time to subsist. It is for the last something of the [234a]

γεγονότος, οὗ ἐπὶ τάδε οὐθέν ἐστι τοῦ μέλλοντος, καὶ

of what has happened, of which upon these nothing is of the future, and

πάλιν τοῦ μέλλοντος, οὗ ἐπὶ τάδε οὐθέν ἐστι τοῦ γεγονότος·

again of the future, of which upon these nothing is of the of what has happened;

ὃ δή φαμεν ἀμφοῖν εἶναι πέρας. τοῦτο δὲ ἐὰν δειχθῇ ὅτι

which indeed we say of both to be the end. This but if it is shown that

τοιοῦτόν ἐστιν [καθ' αὐτὸ] καὶ ταὐτόν, ἅμα φανερὸν

such is [according to itself] and the same, at the same time it will be evident

ἔσται καὶ ὅτι ἀδιαίρετον. ἀνάγκη δὴ τὸ αὐτὸ εἶναι τὸ νῦν

it will be and that indivisible. Necessity indeed the same to be the now

τὸ ἔσχατον ἀμφοτέ ρων τῶν χρόνων· εἰ γὰρ ἕτερον, ἐφεξῆς

the last both of the times. If for another, subsequently

μὲν οὐκ ἂν εἴη θάτερον θατέρῳ διὰ τὸ μὴ εἶναι συνεχὲς

indeed not would be another to the other through the not to be continuous

ἐξ ἀμερῶν, εἰ δὲ χωρὶς ἑκάτερον, μεταξὺ ἔσται χρόνος·

from parts, if but separately each, in between there will be time.

πᾶν γὰρ τὸ συνεχὲς τοιοῦτον ὥστ' εἶναί τι συνώνυμον μεταξὺ

all for the continuous such that it is something synonymous between

τῶν περάτων. ἀλλὰ μὴν εἰ χρόνος τὸ μεταξύ, διαιρετὸν ἔσται·

the limits. but indeed if time the in between, divisible it will be.

πᾶς γὰρ χρόνος δέδεικται ὅτι διαιρετός. ὥστε διαιρετὸν τὸ

all for time has been shown that it is divisible. therefore divisible the

νῦν. εἰ δὲ διαιρετὸν τὸ νῦν, ἔσται τι τοῦ γεγονότος ἐν

now. If but divisible the now, there will be something of the event in

τῷ μέλλοντι καὶ τοῦ μέλλοντος ἐν τῷ γεγονότι· καθ' ὃ

the future and of the future in the event; according to what

γὰρ ἂν διαιρεθῇ, τοῦτο διοριεῖ τὸν παρήκοντα καὶ τὸν μέλλοντα

for would is divided, this will define the passing and the future

χρόνον. ἅμα δὲ καὶ οὐκ ἂν καθ' αὐτὸ εἴη τὸ νῦν,

time. at the same time but and not would according to itself be the now,

ἀλλὰ καθ' ἕτερον· ἡ γὰρ διαίρεσις οὐ καθ' αὐτό.

but according to another; the for division is not according to itself.

πρὸς δὲ τούτοις τοῦ νῦν τὸ μέν τι γεγονὸς

in relation to but these of the now the on the one hand something having happened

ἔσται τὸ δὲ μέλλον, καὶ οὐκ ἀεὶ τὸ αὐτὸ γεγονὸς ἢ μέλλον.

will be the but future, and not always the same having happened or future.

οὐδὲ δὴ τὸ νῦν τὸ αὐτό· πολλαχῇ γὰρ διαιρετὸς ὁ χρόνος.

neither indeed the now the same; in many ways for divisible the time.

ὥστ' εἰ ταῦτα ἀδύνατον ὑπάρχειν, ἀνάγκη τὸ αὐτὸ εἶναι τὸ ἐν

thus if these impossible to exist, necessity the same to be the in

ἑκατέρῳ νῦν. ἀλλὰ μὴν εἰ ταὐτό, φανερὸν ὅτι καὶ ἀδιαίρετον·

each now. but indeed if it is the same, evident that and indivisible·

εἰ γὰρ διαιρετόν, πάλιν ταὐτὰ συμβήσεται ἃ καὶ ἐν τῷ πρότερον.

if for divisible, again the same will happen which and in the before.

ὅτι μὲν τοίνυν ἔστιν τι ἐν τῷ χρόνῳ ἀδιαίρετον, ὃ

that on one hand therefore there is something in the time indivisible, which

φαμεν εἶναι τὸ νῦν, δῆλόν ἐστιν ἐκ τῶν εἰρημένων·

we say to be the now, clear is from the aforementioned·

ὅτι δ' οὐθὲν ἐν τῷ νῦν κινεῖται, ἐκ τῶνδε φανερόν ἐστιν. εἰ

that but nothing in the now is moving, from these clear is. if

γάρ, ἐνδέχεται καὶ θᾶττον κινεῖσθαι καὶ βραδύτερον. ἔστω

indeed, it is possible also quicker to be moved and slower. let it be

δὴ τὸ νῦν ἐφ' ᾧ N, κεκινήσθω δ' ἐν αὐτῷ τὸ θᾶττον

indeed the now regarding which N, let it start but in it the quicker

τὴν AB. οὐκοῦν τὸ βραδύτερον ἐν τῷ αὐτῷ ἐλάττω τῆς AB

the AB. therefore the slower in the same less of the AB

κινηθήσεται, οἷον τὴν AΓ. ἐπεὶ δὲ τὸ βραδύτερον ἐν ὅλῳ τῷ νῦν

will be moved, such as the AG. since but the slower in all the now

κεκίνηται τὴν AΓ, τὸ θᾶττον ἐν ἐλάττονι τούτου κινηθήσεται,

has been moved the AG, the quicker in less than this will be moved,

ὥστε διαιρεθήσεται τὸ νῦν. ἀλλ' ἦν ἀδιαίρετον. οὐκ ἄρα ἔστιν

so that it will be divided the now. but it was indivisible. not therefore it is

κινεῖσθαι ἐν τῷ νῦν.

to be moved in the now.

ἀλλὰ μὴν οὐδ' ἠρεμεῖν· ἠρεμεῖν γὰρ λέγομεν τὸ πεφυκὸς

but indeed nor to be at rest; to be at rest for we say the natural

κινεῖσθαι μὴ κινούμενον ὅτε πέφυκεν καὶ οὗ καὶ ὡς, ὥστ' ἐπεὶ

to be moved not being moved when it has become and of which and how, so that since

ἐν τῷ νῦν οὐθὲν πέφυκε κινεῖσθαι, δῆλον ὡς οὐδ' ἠρεμεῖν.

in the present nothing has become to be moved, clear that not even to be at rest.

ἔτι δ' εἰ τὸ αὐτὸ μέν ἐστι τὸ νῦν ἐν ἀμφοῖν τοῖν χρόνοιν,

also but if the same on one hand is the present in both the times,

[234b] ἐνδέχεται δὲ τὸν μὲν κινεῖσθαι τὸν δ'

[234b] it may happen but the on one hand to be moved the on the other hand

ἠρεμεῖν ὅλον, τὸ δ' ὅλον κινούμενον τὸν χρόνον ἐν

to be at rest altogether, the on the other hand all being moved the time in

ὁτῳοῦν κινηθήσεται τῶν τούτου καθ' ὃ πέφυκε κινεῖσθαι,

whatever will be moved of those this according to that has become to be moved,

καὶ τὸ ἠρεμοῦν ὡσαύτως ἠρεμήσει, συμβήσεται τὸ αὐτὸ ἅμα

and the being at rest likewise will be at rest, will happen the same together

ἠρεμεῖν καὶ κινεῖσθαι· τὸ γὰρ αὐτὸ ἔσχατον τῶν χρόνων
to be at rest and to be moved; the for the same last of the times
ἀμφοτέρων, τὸ νῦν.
both, the now.

ἔτι δ' ἠρεμεῖν μὲν λέγομεν τὸ ὁμοίως ἔχον καὶ αὐτὸ καὶ τὰ
still but to be at rest on one hand we say the likewise having and it and the
μέρη νῦν καὶ πρότερον· ἐν δὲ τῷ νῦν οὐκ ἔστι τὸ πρότερον, ὥστ'
parts now and earlier in but the now not is the earlier, thus
οὐδ' ἠρεμεῖν.
neither to be at rest.

ἀνάγκη ἄρα καὶ κινεῖσθαι τὸ κινούμενον ἐν χρόνῳ καὶ ἠρεμεῖν
necessity therefore and to be moved the moved in time and to be at rest
τὸ ἠρεμοῦν.
the resting.

Κεφάλαιο 4

Chapter 4

Τὸ δὲ μεταβάλλον ἅπαν ἀνάγκη διαιρετὸν εἶναι. ἐπεὶ γὰρ ἔκ
The but changing all necessity to be divisible to be. since indeed from
τινος εἴς τι πᾶσα μεταβολή, καὶ ὅταν μὲν ᾖ ἐν
a certain into something all change, and whenever on the one hand it is in
τούτῳ εἰς ὃ μετέβαλλεν, οὐκέτι μεταβάλλει, ὅταν δὲ ἐξ οὗ
this towards that it has changed, no longer changes, whenever but out of which
μετέβαλλεν, καὶ αὐτὸ καὶ τὰ μέρη πάντα, οὔπω μεταβάλλει (τὸ γὰρ
it has changed, and it and the parts all, not yet it changes (the for

ὡσαύτως ἔχον καὶ αὐτὸ καὶ τὰ μέρη οὐ μεταβάλλει), ἀνάγκη οὖν

likewise having and it and the parts not changes), necessity therefore

τὸ μέν τι ἐν τούτῳ εἶναι, τὸ δ' ἐν θατέρῳ

the on the one hand something in this to be, the on the other hand in another

τοῦ μεταβάλλοντος· οὔτε γὰρ ἐν ἀμφοτέροις οὔτ' ἐν μηδετέρῳ

of the changing· neither for in both nor in either

δυνατόν. λέγω δ' εἰς ὃ μεταβάλλει τὸ πρῶτον κατὰ τὴν

possible. I say but into what changes the first according to the

μεταβολήν, οἷον ἐκ τοῦ λευκοῦ τὸ φαιόν, οὐ τὸ μέλαν· οὐ γὰρ

change, such as from the white the gray, not the black· not for

ἀνάγκη τὸ μεταβάλλον ἐν ὁποτερῳοῦν εἶναι τῶν ἄκρων. φανερὸν

necessity to be changing in whichever to be of the extremes. evident

οὖν ὅτι πᾶν τὸ μεταβάλλον ἔσται διαιρετόν.

therefore that everything the changing will be divisible.

κίνησις δ' ἐστὶν διαιρετὴ διχῶς, ἕνα μὲν τρόπον τῷ

motion but is divisible in two ways, one on the one hand way in the

χρόνῳ, ἄλλον δὲ κατὰ τὰς τῶν μερῶν τοῦ κινουμένου

time, another on the other hand according to the of the parts of the moving

κινήσεις, οἷον εἰ τὸ ΑΓ κινεῖται ὅλον, καὶ τὸ ΑΒ κινήσεται καὶ

motions, such as if the AG is moved wholly, and the AB will be moved and

τὸ ΒΓ. ἔστω δὴ τοῦ μὲν ΑΒ ἡ ΔΕ, τοῦ δὲ ΒΓ

the BC. let it be indeed of the on the one hand AB the DE, of the but BC

ἡ ΕΖ κίνησις τῶν μερῶν. ἀνάγκη δὴ τὴν ὅλην, ἐφ' ἧς ΔΖ,

the EF motion of the parts. necessity indeed the whole, upon which DZ,

τοῦ ΑΓ εἶναι κίνησιν. κινήσεται γὰρ κατὰ ταύτην, ἐπείπερ

of the AG to be a motion. will be moved for according to this, since

ἑκάτερον τῶν μερῶν κινεῖται καθ' ἑκατέραν· οὐθὲν δὲ κινεῖται
each of the parts is moved according to each part; nothing but is moved
κατὰ τὴν ἄλλου κίνησιν· ὥστε ἡ ὅλη κίνησις τοῦ ὅλου
according to the other motion; so that the whole motion of the of the whole
ἐστὶν μεγέθους κίνησις.
is of magnitude motion.

ἔτι δ' εἰ πᾶσα μὲν κίνησις τινός, ἡ δ' ὅλη κίνησις ἡ
Moreover and if every indeed motion of some, the and whole motion the
ἐφ' ἧς ΔΖ μήτε τῶν μερῶν ἐστιν μηδετέρου (μέρους γὰρ
depending on which DZ neither of the parts exists of neither (part for
ἑκατέρα) μήτ' ἄλλου μηδενός (οὗ γὰρ ὅλη ὅλου, καὶ τὰ
each) nor of another of none (of which for the whole of the whole, and the
μέρη τῶν μερῶν· τὰ δὲ μέρη τῶν ΑΒ ΒΓ καὶ οὐδένων ἄλλων·
parts of the parts; the but parts of the AB BG and of none; of others;
πλειόνων γὰρ οὐκ ἦν μία κίνησις), κἂν ἡ ὅλη κίνησις εἴη
of more for not was one motion), and even the whole motion might be
τοῦ ΑΒΓ μεγέθους.
of the ABC of magnitude.

ἔτι δ' εἰ ἔστιν ἄλλη τοῦ ὅλου κίνησις, οἷον ἐφ' ἧς ΘΙ,
still and if there is another of the whole motion, such as concerning which THI,
ἀφαιρεθήσεται ἀπ' αὐτῆς [235b] ἡ ἑκατέρων τῶν μερῶν κίνησις·
will be taken away from it [235b] the of each of the parts motion·
αὗται δ' ἴσαι ἔσονται ταῖς ΔΕ ΕΖ· μία γὰρ ἑνὸς κίνησις. ὥστ'
these and equal will be to the DE EZ· one for of one motion. thus
εἰ μὲν ὅλη διαιρεθήσεται ἡ ΘΙ εἰς τὰς τῶν μερῶν
if on the one hand the whole will be divided the THI into the of the parts

κινήσεις, ἴση ἔσται ἡ ΘΙ τῇ ΔΖ· εἰ δ' ἀπολείπει τι, οἷον

motions, equal will be the THI to the DZ· if and leaves out something, such as

τὸ ΚΙ, αὕτη οὐδενὸς ἔσται κίνησις (οὔτε γὰρ τοῦ ὅλου οὔτε

the KI, this of no one will be motion (neither for of the whole nor

τῶν μερῶν διὰ τὸ μίαν εἶναι ἑνός, οὔτε ἄλλου οὐθενός· ἡ

of the parts through the one to be of one, nor of another of nothing· the

γὰρ συνεχὴς κίνησίς ἐστι συνεχῶν τινῶν), ὡσαύτως δὲ καὶ

for continuous motion is of those continuing certain), likewise and also

εἰ ὑπερβάλλει κατὰ τὴν διαίρεσιν· ὥστ' εἰ τοῦτο ἀδύνατον,

if it surpasses according to the division; so that if this is impossible,

ἀνάγκη τὴν αὐτὴν εἶναι καὶ ἴσην.

necessity the same to be and equal.

αὕτη μὲν οὖν ἡ διαίρεσις κατὰ τὰς τῶν μερῶν

this on the one hand therefore the division according to the of the parts

κινήσεις ἐστίν, καὶ ἀνάγκη παντὸς εἶναι τοῦ μεριστοῦ αὐτήν·

motions is, and necessity of everything to be of the divisible this;

ἄλλη δὲ κατὰ τὸν χρόνον· ἐπεὶ γὰρ ἅπασα κίνησις ἐν χρόνῳ,

another but according to the time; since for all motion in time,

χρόνος δὲ πᾶς διαιρετός, ἐν δὲ τῷ ἐλάττονι ἐλάττων ἡ κίνησις,

time but every divisible, in the the lesser lesser the motion,

ἀνάγκη πᾶσαν κίνησιν διαιρεῖσθαι κατὰ τὸν χρόνον. ἐπεὶ δὲ

necessity of every motion to be divided according to the time. since but

πᾶν τὸ (13) κινούμενον ἔν τινι κινεῖται καὶ χρόνον τινά,

everything the (13) moving in some is moved and time some time,

καὶ παντὸς ἔστι κίνησις, ἀνάγκη τὰς αὐτὰς εἶναι διαιρέσεις

and of everything there is motion, necessity the same to be divisions

τοῦ τε χρόνου καὶ τῆς κινήσεως καὶ τοῦ κινεῖσθαι καὶ τοῦ

of the both time and of the motion and of the being moved and of the

κινουμένου καὶ ἐν ᾧ ἡ κίνησις (πλὴν οὐ πάντων ὁμοίως ἐν οἷς

moving and in which the movement (except not all similarly in which

ἡ κίνησις, ἀλλὰ τοῦ μὲν τόπου καθ' αὑτό, τοῦ δὲ ποιοῦ

the movement, but of the one place according to itself, of the and of the doer

κατὰ συμβεβηκός). εἰλήφθω γὰρ ὁ χρόνος ἐν ᾧ κινεῖται

according to accident). let it be taken for the time in which it is moved

ἐφ' ᾧ A, καὶ ἡ κίνησις ἐφ' ᾧ B. εἰ οὖν τὴν ὅλην ἐν τῷ

upon which A, and the movement upon which B. if therefore the whole in the

παντὶ χρόνῳ κεκίνηται, ἐν τῷ ἡμίσει ἐλάττω, καὶ πάλιν τούτου

all time is moved, in the half less, and again of this

διαιρεθέντος ἐλάττω ταύτης, καὶ ἀεὶ οὕτως.

having been divided less of this, and always thus.

ὁμοίως δὲ καί, εἰ ἡ κίνησις διαιρετή, καὶ ὁ χρόνος διαιρετός· εἰ

likewise but and, if the motion divisible, and the time is divisible. if

γὰρ τὴν ὅλην ἐν τῷ παντί, τὴν ἡμίσειαν ἐν τῷ ἡμίσει, καὶ πάλιν τὴν

for the whole in the all, the half in the half, and again the

ἐλάττω ἐν τῷ ἐλάττονι. τὸν αὐτὸν δὲ τρόπον καὶ τὸ κινεῖσθαι

lesser in the lesser. the same but manner and the to be moved

διαιρεθήσεται. ἔστω γὰρ ἐφ' ᾧ Γ τὸ κινεῖσθαι. κατὰ

will be divided. let it be for on which G the moving. according to

δὴ τὴν ἡμίσειαν κίνησιν ἔλαττον ἔσται τοῦ ὅλου, καὶ πάλιν

indeed the half motion lesser will be of the whole, and again

κατὰ τὴν τῆς ἡμισείας ἡμίσειαν, καὶ αἰεὶ οὕτως. ἔστι δὲ

according to the of the half half, and always thus. there is but

καὶ ἐκθέμενον τὸ καθ' ἑκατέραν τῶν κινήσεων κινεῖσθαι,

and setting forth the according to each of the motions to be moved,

οἷον κατά τε τὴν ΔΓ καὶ τὴν ΓΕ, λέγειν ὅτι τὸ ὅλον ἔσται

such as according to both the DG and the O to say that the whole will be

κατὰ τὴν ὅλην (εἰ γὰρ ἄλλο, πλείω ἔσται κινεῖσθαι κατὰ

according to the total (if for another more will be to be moved according to

τὴν αὐτὴν κίνησιν), ὥσπερ ἐδείξαμεν καὶ τὴν κίνησιν διαιρετὴν εἰς

the same movement), just as we have shown and the movement divisible into

τὰς τῶν μερῶν κινήσεις οὖσαν· ληφθέντος γὰρ τοῦ κινεῖσθαι

the of the parts movements being· having been taken for the to be moved

καθ' ἑκατέραν συνεχὲς ἔσται τὸ ὅλον.

in accordance with each continuous will be the whole.

ὡσαύτως δὲ δειχθήσεται καὶ τὸ μῆκος διαιρετόν, καὶ ὅλως πᾶν

similarly but it will be shown and the length divisible, and altogether all

ἐν ᾧ ἐστιν ἡ μεταβολή (πλὴν ἔνια κατὰ συμβεβηκός, ὅτι

in which is the change (except some according to accident, that

τὸ μεταβάλλον ἐστὶν διαιρετόν)· ἑνὸς γὰρ διαιρουμένου πάντα

the changing is divisible). of one for being divided all

διαιρεθήσεται. καὶ ἐπὶ τοῦ [235c] πεπερασμένα εἶναι ἢ ἄπειρα ὁμοίως

will be divided. And upon the [235c] finite to be or infinite similarly

ἕξει κατὰ πάντων. ἠκολούθηκεν δὲ μάλιστα τὸ διαιρεῖσθαι

will have according to all. It has followed but especially the to be divided

πάντα καὶ ἄπειρα εἶναι ἀπὸ τοῦ μεταβάλλοντος· εὐθὺς γὰρ

all and infinite to be from the changing. Immediately for

ἐνυπάρχει τῷ μεταβάλλοντι τὸ διαιρετὸν καὶ τὸ ἄπειρον. τὸ μὲν

it exists in the changing the divisible and the infinite. The indeed

οὖν διαιρετὸν δέδεικται πρότερον, τὸ δ' ἄπειρον ἐν τοῖς

therefore the divisible has been shown previously, the but infinite in the

ἑπομένοις ἔσται δῆλον.

following will be clear.

Κεφάλαιο 5

Chapter 5

Ἐπεὶ δὲ πᾶν τὸ μεταβάλλον ἔκ τινος εἴς τι μεταβάλλει,

Since but all the changing from something into something it changes,

ἀνάγκη τὸ μεταβεβληκός, ὅτε πρῶτον μεταβέβληκεν, εἶναι ἐν

necessity the changed, when first it has changed, to be in

ᾧ μεταβέβληκεν. τὸ γὰρ μεταβάλλον, ἐξ οὗ μεταβάλλει,

which it has changed. the for changing, out of which it changes,

ἐξίσταται ἢ ἀπολείπει αὐτό, καὶ ἤτοι ταὐτόν ἐστι τὸ μεταβάλλειν

stands out or leaves it, and either the same is the changing

καὶ τὸ ἀπολείπειν, ἢ ἀκολουθεῖ τῷ μετα βάλλειν τὸ ἀπολείπειν.

and the leaving, or it follows the changing to change the leaving.

εἰ δὲ τῷ μεταβάλλειν τὸ ἀπολείπειν, τῷ μεταβεβληκέναι τὸ

if but to the changing the leaving, to the having changed the

ἀπολελοιπέναι· ὁμοίως γὰρ ἑκάτερον ἔχει πρὸς ἑκάτερον.

having left; similarly for each one has towards each one.

ἐπεὶ οὖν μία τῶν μεταβολῶν ἡ κατ' ἀντίφασιν, ὅτε μεταβέβληκεν

since therefore one of the changes the according to contradiction, when it has changed

ἐκ τοῦ μὴ ὄντος εἰς τὸ ὄν, ἀπολέλοιπεν τὸ μὴ ὄν. ἔσται

from the non- being into the being, it has left behind the non- being. it will be

ἄρα ἐν τῷ ὄντι· πᾶν γὰρ ἀνάγκη ἢ εἶναι ἢ μὴ εἶναι.

thus in the being; everything indeed necessity either to be or not to be.

φανερὸν οὖν ὅτι ἐν τῇ κατ' ἀντίφασιν μεταβολῇ τὸ μεταβεβληκὸς
evident therefore that in the according to contradiction change the changed
ἔσται ἐν ᾧ μεταβέβληκεν. εἰ δ' ἐν ταύτῃ, καὶ ἐν ταῖς ἄλλαις·
will be in which it has changed. if however in this, and in the others;
ὁμοίως γὰρ ἐπὶ μιᾶς καὶ τῶν ἄλλων.
similarly indeed upon one and of the others.

ἔτι δὲ καὶ καθ' ἑκάστην λαμβάνουσι φανερόν, εἴπερ ἀνάγκη
still but and according to each take evident, if indeed necessity
τὸ μεταβεβληκὸς εἶναί που ἢ ἔν τινι. ἐπεὶ γὰρ ἐξ οὗ
the transformed to be somewhere or in something. since indeed from which
μεταβέβληκεν ἀπολέλοιπεν, ἀνάγκη δ' εἶναί που, ἢ ἐν τούτῳ
it has transformed it has left, necessity but to be somewhere, or in this
ἢ ἐν ἄλλῳ ἔσται. εἰ μὲν οὖν ἐν ἄλλῳ, οἷον ἐν τῷ
or in another will be. if on the one hand therefore in another, such as in the
Γ, τὸ εἰς τὸ Β μεταβεβληκός, πάλιν ἐκ τοῦ Γ μεταβάλλει εἰς
G, the into the B transformed, again from the G changes into
τὸ Β· οὐ γὰρ ἦν ἐχόμενον τὸ Β, ἡ δὲ μεταβολὴ συνεχής.
the B. not indeed was holding the B, the and transformation continuous.
ὥστε τὸ μεταβεβληκός, ὅτε μεταβέβληκεν, μεταβάλλει εἰς ὃ
so the transformed, when it has transformed, it changes into whatever
μεταβέβληκεν. τοῦτο δ' ἀδύνατον· ἀνάγκη ἄρα τὸ μεταβεβληκὸς
it has transformed. this but impossible; necessity therefore the transformed
εἶναι ἐν τούτῳ εἰς ὃ μεταβέβληκεν. φανερὸν οὖν ὅτι καὶ τὸ
to be in this into which it has changed. evident therefore that and the
γεγονός, ὅτε γέγονεν, ἔσται, καὶ τὸ ἐφθαρμένον οὐκ ἔσται·
fact, when it has come to be, it will be, and the corrupted not will be.

καθόλου τε γὰρ εἴρηται περὶ πάσης μεταβολῆς, καὶ μάλιστα

in general and for it has been said about all change, and especially

δῆλον ἐν τῇ κατ' ἀντίφασιν. ὅτι μὲν τοίνυν τὸ μεταβεβληκός,

clear in the according to contradiction. that indeed therefore the changed one,

ὅτε μεταβέβληκε πρῶτον, ἐν ἐκείνῳ ἐστίν, δῆλον·

when it has changed first, in that it is, clear.

ἐν ᾧ δὲ πρώτῳ μεταβέβληκεν τὸ μεταβεβληκός, ἀνάγκη ἄτομον

in which but first has changed the changed necessity atom

εἶναι. λέγω δὲ πρῶτον ὃ μὴ τῷ ἕτερόν τι αὐτοῦ εἶναι

to be. I say but first that not to the other something of it to be

τοιοῦτόν ἐστιν. ἔστω γὰρ διαιρετὸν τὸ ΑΓ, καὶ διῃρήσθω

such it is. let it be for divisible the AG, and let it be distinguished

κατὰ τὸ Β. εἰ μὲν οὖν ἐν τῷ ΑΒ μετα βέβληκεν ἢ

according to the B. if indeed therefore in the AB after has been changed or

πάλιν ἐν τῷ ΒΓ, οὐκ ἂν ἐν πρώτῳ τῷ ΑΓ με ταβεβληκὸς εἴη.

again in the BG, not would in the first the AG me changed would be.

εἰ δ' ἐν ἑκατέρῳ μετέβαλλεν (ἀνάγκη γὰρ [236a] ἢ μεταβεβληκέναι

if but in each it changed (necessity for [236a] or to have changed

ἢ μεταβάλλειν ἐν ἑκατέρῳ), κἂν ἐν τῷ ὅλῳ μεταβάλλοι· ἀλλ'

or to change in each), and if in the whole it would change; but

ἦν μεταβεβληκός. ὁ αὐτὸς δὲ λόγος καὶ εἰ ἐν τῷ μὲν

it was changed. the same but reason and if in the on the one hand

μεταβάλλει, ἐν δὲ τῷ μεταβέβληκεν· ἔσται γάρ τι τοῦ

changes, in but the has changed. it will be for something of the

πρώτου πρότερον· ὥστ' οὐκ ἂν εἴη διαιρετὸν ἐν ᾧ

first earlier; so that not at all it would be divisible in which

μεταβέβληκεν. φανερὸν οὖν ὅτι καὶ τὸ ἐφθαρμένον καὶ τὸ

it has changed. evident therefore that both the perishable and the

γεγονὸς ἐν ἀτόμῳ τὸ μὲν ἔφθαρται τὸ δὲ γέγονεν.

generated in the atom the on the one hand is perishable the but has come to be.

λέγεται δὲ τὸ ἐν ᾧ πρώτῳ μεταβέβληκε διχῶς, τὸ μὲν

it is said but the in which first has changed differently, the on one hand

ἐν ᾧ πρώτῳ ἐπετελέσθη ἡ μεταβολή (τότε γὰρ ἀληθὲς εἰπεῖν

in which first was completed the change (then for true to say

ὅτι μεταβέβληκεν), τὸ δ' ἐν ᾧ πρώτῳ ἤρξατο μεταβάλλειν.

that it has changed), the and in which first began to change.

τὸ μὲν οὖν κατὰ τὸ τέλος τῆς μεταβολῆς πρῶτον

the on one hand therefore according to the end of the change first

λεγόμενον ὑπάρχει τε καὶ ἔστιν (ἐνδέχεται γὰρ ἐπιτελεσθῆναι

called exists and also is (it is possible for to be completed

μεταβολὴν καὶ ἔστι μεταβολῆς τέλος, ὃ δὴ καὶ δέδεικται

change and it is of change end, which indeed and has been demonstrated

ἀδιαίρετον ὂν διὰ τὸ πέρας εἶναι)· τὸ δὲ κατὰ τὴν ἀρχὴν

indivisible being through the limit to be)· the but according to the beginning

ὅλως οὐκ ἔστιν· οὐ γὰρ ἔστιν ἀρχὴ μεταβολῆς, οὐδ' ἐν ᾧ

altogether not is; not for there is a beginning of change, nor in which

πρώτῳ τοῦ χρόνου μετέβαλλεν. ἔστω γὰρ πρῶτον ἐφ' ᾧ τὸ

first of the time it changed. let it be for first on which the

ΑΔ. τοῦτο δὴ ἀδιαίρετον μὲν οὐκ ἔστιν· συμβήσεται γὰρ

AD. this indeed indivisible on the one hand not is. it will occur for

ἐχόμενα εἶναι τὰ νῦν. ἔτι δ' εἰ ἐν τῷ ΓΑ χρόνῳ παντὶ ἠρεμεῖ

being to be the now. still but if in the GA time all it is at rest

(κείσθω γὰρ ἠρεμοῦν), καὶ ἐν τῷ Α ἠρεμεῖ, ὥστ' εἰ ἀμερές

(let it lie for being at rest), and in the A is at rest, so that if indivisible

ἐστι τὸ ΑΔ, ἅμα ἠρεμήσει καὶ μεταβεβληκὸς ἔσται·

it is the AD, together it will also be at rest and having been changed it will be.

ἐν μὲν γὰρ τῷ Α ἠρεμεῖ, ἐν δὲ τῷ Δ μεταβέβληκεν.

in on the one hand for the A is at rest, in but the D it has changed.

ἐπεὶ δ' οὐκ ἔστιν ἀμερές, ἀνάγκη διαιρετὸν εἶναι καὶ ἐν

since but not it is indivisible, necessity to be divisible to be and in

ὁτῳοῦν τῶν τούτου μεταβεβληκέναι· διαιρεθέντος γὰρ τοῦ

any of the this to have changed; having been divided for of the

ΑΔ, εἰ μὲν ἐν μηδετέρῳ μεταβέβληκεν, οὐδ' ἐν τῷ ὅλῳ·

AD, if on the one hand in neither has changed, nor in the whole.

εἰ δ' ἐν ἀμφοῖν μεταβάλλει καὶ ἐν τῷ παντί, εἴτ' ἐν θατέρῳ

if but in both changes and in the all, whether in the other

μεταβέβληκεν, οὐκ ἐν τῷ ὅλῳ πρώτῳ. ὥστε ἀνάγκη ἐν ὁτῳοῦν

has changed, not in the whole first. thus necessity in whatever

μεταβεβλη κέναι. φανερὸν τοίνυν ὅτι οὐκ ἔστιν ἐν ᾧ πρώτῳ

has changed to be absent. evident therefore that not is in which first

μεταβέβληκεν· ἄπειροι γὰρ αἱ διαιρέσεις. οὐδὲ δὴ τοῦ μεταβεβληκότος

has changed. limitless for the divisions. nor indeed of the changed

ἔστιν τι πρῶτον ὃ μεταβέβληκεν. ἔστω γὰρ τὸ ΔΖ πρῶτον

is anything first that has changed. let it be for the DZ first

μεταβεβληκὸς τοῦ ΔΕ· πᾶν γὰρ δέδεικται διαιρετὸν τὸ

having changed of the DE. everything for has been shown to be divisible the

μεταβάλλον. ὁ δὲ χρόνος ἐν ᾧ τὸ ΔΖ μεταβέβληκεν ἔστω ἐφ'

changing. the but time in which the DZ has changed let it be for

ᾧ ΘΙ. εἰ οὖν ἐν τῷ παντὶ τὸ ΔΖ μεταβέβληκεν, ἐν τῷ ἡμίσει

which THI. if then in to the whole the ΔZ has changed, in the half

ἔλαττον ἔσται τι μεταβεβληκὸς καὶ πρότερον τοῦ ΔΖ, καὶ

less it will be something changed and previously of the ΔΖ, and

πάλιν τούτου ἄλλο, κἀκείνου ἕτερον, καὶ αἰεὶ οὕτως. ὥστ' οὐθὲν

again this, another, and that other, and always thus. so that nothing

ἔσται πρῶτον τοῦ μεταβάλλοντος ὃ μεταβέβληκεν. ὅτι μὲν

will be first of the changing which has been changed. that on one hand

οὖν οὔτε τοῦ μεταβάλλοντος οὔτ' ἐν ᾧ μεταβάλλει χρόνῳ

therefore neither of the changing nor in which changes time

πρῶτον οὐθέν ἐστιν, φανερὸν ἐκ τῶν εἰρημένων·

first nothing exists, clear from the aforementioned.

[236b] αὐτὸ δὲ ὃ μεταβάλλει ἢ καθ' ὃ μεταβάλλει, οὐκέθ'

[236b] it but which changes or according to which changes, no longer

ὁμοίως ἕξει. τρία γάρ ἐστιν ἃ λέγεται κατὰ τὴν μεταβολήν,

likewise will have. three for it is that is said according to the change,

τό τε μεταβάλλον καὶ ἐν ᾧ καὶ εἰς ὃ μεταβάλλει, οἷον ὁ

the and changing and in which also into which it changes, such as the

ἄνθρωπος καὶ ὁ χρόνος καὶ τὸ λευκόν. ὁ μὲν οὖν

human and the time and the white. the on one hand therefore

ἄνθρωπος καὶ ὁ χρόνος διαιρετοί, περὶ δὲ τοῦ λευκοῦ ἄλλος

human and the time are divisible, about but the white another

λόγος. πλὴν κατὰ συμβεβηκός γε πάντα διαιρετά· ᾧ

discourse. except according to the accidental indeed all are divisible. which

γὰρ συμβέβηκεν τὸ λευκὸν ἢ τὸ ποιόν, ἐκεῖνο διαιρετόν ἐστιν·

for has happened the white or the quality, that is divisible it is.

ἐπεὶ ὅσα γε καθ' αὐτὰ λέγεται διαιρετὰ καὶ μὴ κατὰ

since all indeed according to themselves is said divisible and not according to

συμβεβηκός, οὐδ' ἐν τούτοις ἔσται τὸ πρῶτον, οἷον ἐν τοῖς

the accidental, not in these will be the first, such as in the

μεγέθεσιν. ἔστω γὰρ τὸ ἐφ' ᾧ AB μέγεθος, κεκινήσθω δ' ἐκ

sizes. let it be for the on which AB size, let it be moved but from

τοῦ B εἰς τὸ Γ πρῶτον. οὐκοῦν εἰ μὲν ἀδιαίρετον ἔσται τὸ

the B to the C first. therefore if on the one hand indivisible will be the

BΓ, ἀμερὲς ἀμεροῦς ἔσται ἐχόμενον· εἰ δὲ διαιρετόν, ἔσται

BC, unseparated of the unseparated will be having, if but divisible, will be

τι τοῦ Γ πρότερον, εἰς ὃ μεταβέβληκεν, κἀκείνου πάλιν

something of the C prior, to what has changed, and from it again

ἄλλο, καὶ ἀεὶ οὕτως διὰ τὸ μηδέποτε ὑπολείπειν τὴν διαίρεσιν.

another, and always thus through the never to leave behind the division.

ὥστ' οὐκ ἔσται πρῶτον εἰς ὃ μεταβέβλη κεν. ὁμοίως δὲ καὶ ἐπὶ

so that not will be first to what has changed empty. similarly but and upon

τῆς τοῦ ποσοῦ μεταβολῆς· καὶ γὰρ αὕτη ἐν συνεχεῖ ἐστιν.

the of the quantity change; and for this in it is continuous is.

φανερὸν οὖν ὅτι ἐν μόνῃ τῶν κινήσεων τῇ κατὰ τὸ

obvious therefore that in the single of the movements the according to the

ποιὸν ἐνδέχεται ἀδιαίρετον καθ' αὐτὸ εἶναι.

quality it is possible indivisible throughout itself to be.

Κεφάλαιο 6

Chapter 6

Ἐπεὶ δὲ τὸ μεταβάλλον ἅπαν ἐν χρόνῳ μεταβάλλει, λέγεται δ' ἐν

Since but the changing all in time changes, it is said that in

χρόνῳ μεταβάλλειν καὶ ὡς ἐν πρώτῳ καὶ ὡς καθ' ἕτερον, οἷον

time to change and as in the first and as according to another, such as

ἐν τῷ ἐνιαυτῷ ὅτι ἐν τῇ ἡμέρᾳ μεταβάλλει, ἐν ᾧ πρώτῳ χρόνῳ

in the year that in the day changes, in which first time

μεταβάλλει τὸ μεταβάλλον, ἐν ὁτῳοῦν ἀνάγκη τούτου μεταβάλλειν.

it changes the changing, in whatever necessity of this to change.

δῆλον μὲν οὖν καὶ ἐκ τοῦ ὁρισμοῦ (τὸ γὰρ πρῶτον οὕτως

It is clear indeed therefore also from the definition (the for first thus

ἐλέγομεν), οὐ μὴν ἀλλὰ καὶ ἐκ τῶνδε φανερόν. ἔστω γὰρ ἐν

we say), not however but also from these evident. Let it be for in

ᾧ πρώτῳ κινεῖται τὸ κινούμενον ἐφ' ᾧ XP, καὶ διῃρήσθω

which first it is moved the moving on which CR, and let it be divided

κατὰ τὸ K· πᾶς γὰρ χρόνος διαιρετός. ἐν δὴ τῷ XK χρόνῳ

according to the K· every for time is divisible. in indeed the XK time

ἤτοι κινεῖται ἢ οὐ κινεῖται, καὶ πάλιν ἐν τῷ KP ὡσαύτως. εἰ μὲν

or is moved or not is moved, and again in the KR likewise. if indeed

οὖν ἐν μηδετέρῳ κινεῖται, ἠρεμοίη ἂν ἐν τῷ παντί (κινεῖσθαι

then in neither is moved, it would be at rest would in the whole (to be moved

γὰρ ἐν μηθενὶ τῶν τούτου κινούμενον ἀδύνατον)· εἰ δ' ἐν

for in no of the these being moved impossible). if however in

θατέρῳ μόνῳ κινεῖται, οὐκ ἂν ἐν πρώτῳ κινοῖτο τῷ XP·

another alone is moved, not would in the first be moved by the XR.

καθ' ἕτερον γὰρ ἡ κίνησις. ἀνάγκη ἄρα ἐν ὁτῳοῦν τοῦ

according to another for the motion. necessity therefore in any of the

XP κινεῖσθαι.

XR to be moved.

δεδειγμένου δὲ τούτου φανερὸν ὅτι πᾶν τὸ κινούμενον

having been demonstrated but of this it is evident that every the moving

ἀνάγκη κεκινῆσθαι πρότερον. εἰ γὰρ ἐν τῷ ΧΡ πρώτῳ

necessity to have been set in motion previously. if for in the time first

χρόνῳ τὸ ΚΛ κεκίνηται μέγεθος, ἐν τῷ ἡμίσει τὸ ὁμοταχῶς

period the K is moved magnitude, in the half the similarly

κινούμενον καὶ ἅμα ἀρξάμενον τὸ ἥμισυ ἔσται κεκινημένον. εἰ

moving and together having begun the half will be having been moved. if

δὲ τὸ [237a] ὁμοταχὲς ἐν τῷ αὐτῷ χρόνῳ κεκίνηταί τι,

but the [237a] similar in the same time is set in motion something,

καὶ θάτερον ἀνάγκη ταὐτὸ κεκινῆσθαι μέγεθος, ὥστε

and another necessity the same to have been set in motion magnitude, so that

κεκινημένον ἔσται τὸ κινούμενον.

having been moved will be the moving.

ἔτι δὲ εἰ ἐν τῷ παντὶ χρόνῳ τῷ ΧΡ κεκινῆσθαι λέγομεν,

also but if in the all time the time to have been set in motion we say,

ἢ ὅλως ἐν ὁτῳοῦν χρόνῳ, τῷ λαβεῖν τὸ ἔσχατον αὐτοῦ νῦν

or altogether in any time, the taking the ultimate of it now

(τοῦτο γάρ ἐστι τὸ ὁρίζον, καὶ τὸ μεταξὺ τῶν νῦν χρόνος), κἂν

(this for is the horizon, and the between the now time), even if

ἐν τοῖς ἄλλοις ὁμοίως λέγοιτο κεκινῆσθαι. τοῦ δ' ἡμίσεος

in the others likewise it were said to have moved. of the but half

ἔσχατον ἡ διαίρεσις. ὥστε καὶ ἐν τῷ ἡμίσει κεκινημένον ἔσται καὶ

the extreme the division. so that also in the half having been moved will be and

ὅλως ἐν ὁτῳοῦν τῶν μερῶν· ἀεὶ γὰρ ἅμα τῇ τομῇ

altogether in whichever of the parts; always for at the same time the cutting

χρόνος ἔστιν ὡρισμένος ὑπὸ τῶν νῦν. εἰ οὖν ἅπας μὲν

time is defined by the now. if then all on the one hand

χρόνος διαιρετός, τὸ δὲ μεταξὺ τῶν νῦν χρό νος, ἅπαν τὸ μεταβάλλον

time is divisible, the but between the now time is, all the changing

ἄπειρα ἔσται μεταβεβληκός.

infinite will be having changed.

ἔτι δ' εἰ τὸ συνεχῶς μεταβάλλον καὶ μὴ φθαρὲν μηδὲ πεπαυμένον

yet but if the continuously changing and not decayed nor ceased

τῆς μεταβολῆς ἢ μεταβάλλειν ἢ μεταβεβληκέναι ἀναγκαῖον ἐν

of the change or to change or to have changed necessary in

ὁτῳοῦν, ἐν δὲ τῷ νῦν οὐκ ἔστιν μεταβάλλειν, ἀνάγκη μεταβεβληκέναι

any in but the now not is to change, necessity to have changed

καθ' ἕκαστον τῶν νῦν· ὥστ' εἰ τὰ νῦν ἄπειρα, πᾶν τὸ

according to each of the now. so that if the now are infinite, all the

μεταβάλλον ἄπειρα ἔσται μεταβεβληκός.

changing infinite will be having changed.

οὐ μόνον δὲ τὸ μεταβάλλον ἀνάγκη μεταβεβληκέναι, ἀλλὰ καὶ τὸ

not only but the changing necessity to have changed, but also the

μεταβεβληκὸς ἀνάγκη μεταβάλλειν πρότερον· ἅπαν γὰρ τὸ ἔκ

having changed necessity to change previously. all for the from

τινος εἴς τι μεταβεβληκὸς ἐν χρόνῳ μεταβέβληκεν. ἔστω

something into something having changed in time has changed. let it be

γὰρ ἐν τῷ νῦν ἐκ τοῦ A εἰς τὸ B μεταβεβληκός. οὐκοῦν ἐν μὲν

for in the now from the A into the B having changed. therefore in indeed

τῷ αὐτῷ νῦν ἐν ᾧ ἐστιν ἐν τῷ A, οὐ μεταβέβληκεν (ἅμα

the same now in which is in the A, not has changed (at the same time

γὰρ ἂν εἴη ἐν τῷ A καὶ ἐν τῷ B)· τὸ γὰρ μεταβεβληκός,

for would be in the A and in the B)· the for having changed,

ὅτε μεταβέβληκεν, ὅτι οὐκ ἔστιν ἐν τούτῳ, δέδεικται

when it has changed, because it is not is in this, it has been shown

πρότερον· εἰ δ' ἐν ἄλλῳ, μεταξὺ ἔσται χρόνος· οὐ γὰρ ἦν

before· if but in another, in between will be a time· not for it was

ἐχόμενα τὰ νῦν. ἐπεὶ οὖν ἐν χρόνῳ μεταβέβληκεν, χρόνος δ'

having held the now. since therefore in time it has changed, time but

ἅπας διαιρετός, ἐν τῷ ἡμίσει ἄλλο ἔσται μεταβεβληκός, καὶ πάλιν ἐν

all divisible, in the half another will be having changed, and again in

τῷ ἐκείνου ἡμίσει ἄλλο, καὶ αἰεὶ οὕτως· ὥστε μεταβάλλοι ἂν

the that half another, and always thus· so that it would change would

πρότερον.

before.

ἔτι δ' ἐπὶ τοῦ μεγέθους φανερώτερον τὸ λεχθὲν διὰ τὸ συνεχὲς

still but on the magnitude more apparent the said through the continuous

εἶναι τὸ μέγεθος ἐν ᾧ μεταβάλλει τὸ μεταβάλλον. ἔστω γάρ

to be the magnitude in which it changes the changing. let it be for

τι μεταβεβληκὸς ἐκ τοῦ Γ εἰς τὸ Δ. οὐκοῦν εἰ μὲν ἀδιαίρετόν

something changed from the G into the D. therefore if indeed indivisible

ἐστι τὸ ΓΔ, ἀμερὲς ἀμεροῦς ἔσται ἐχόμενον· ἐπεὶ δὲ τοῦτο

is the GD, not separate unseparable it will be holding; since but this

ἀδύνατον, ἀνάγκη μέγεθος εἶναι τὸ μεταξὺ καὶ εἰς ἄπειρα διαιρετόν·

is impossible, necessity magnitude to be the between and into infinite divisible;

ὥστ' εἰς ἐκεῖνα μεταβάλλει πρότερον. ἀνάγκη ἄρα πᾶν τὸ

so that into those it changes beforehand. necessity therefore all the

μεταβεβληκὸς μεταβάλλειν πρότερον. ἡ γὰρ αὐτὴ ἀπόδειξις [237b]

changed to change beforehand. the for itself proof [237b]

καὶ ἐν τοῖς μὴ συνεχέσιν, οἷον ἔν τε τοῖς ἐναντίοις καὶ ἐν

and in the not continuous, such as in both the opposites and in

ἀντιφάσει· ληψόμεθα γὰρ τὸν χρόνον ἐν ᾧ μεταβέβληκεν, καὶ

contradiction. we will take for the time in which it has changed, and

πάλιν ταὐτὰ ἐροῦμεν. ὥστε ἀνάγκη τὸ μεταβεβληκὸς μεταβάλλειν

again the same we will say. therefore necessity the changed to change

καὶ τὸ μεταβάλλον μεταβεβληκέναι, καὶ ἔσται τοῦ μὲν μεταβάλλειν

and the changing to have changed, and it will be of the one changing

τὸ μεταβεβληκέναι πρότερον, τοῦ δὲ μεταβεβληκέναι τὸ μεταβάλλειν,

the having changed before, of the other having changed the changing,

καὶ οὐδέποτε ληφθήσεται τὸ πρῶτον. αἴτιον δὲ τούτου τὸ μὴ εἶναι

and never will it be taken the first. cause but of this the not to be

ἀμερὲς ἀμεροῦς ἐχόμενον· ἄπειρος γὰρ ἡ διαίρεσις, καθάπερ ἐπὶ

without part of a part holding on to; infinite for the division, just as in

τῶν αὐξανομένων καὶ καθαιρουμένων γραμμῶν.

the increasing and decreasing lines.

φανερὸν οὖν ὅτι καὶ τὸ γεγονὸς ἀνάγκη γίγνεσθαι πρότερον

manifest therefore that and the having happened necessity to become previously

καὶ τὸ γιγνό μενον γεγονέναι, ὅσα διαιρετὰ καὶ συνεχῆ, οὐ

and the becoming remain having become, as many as divisible and continuous, not

μέντοι αἰεὶ ὃ γίγνεται, ἀλλ' ἄλλο ἐνίοτε, οἷον τῶν ἐκείνου

however always which becomes, but another sometimes, such as those of that

τι, ὥσπερ τῆς οἰκίας τὸν θεμέλιον. ὁμοίως δὲ καὶ ἐπὶ

something, just as of the house the foundation. similarly and also upon

τοῦ φθειρομένου καὶ ἐφθαρμένου· εὐθὺς γὰρ ἐνυπάρχει τῷ

the being corrupted and having decayed. immediately for it exists in the

γιγνομένῳ καὶ τῷ φθειρομένῳ ἄπειρόν τι συνεχεῖ γε ὄντι,

becoming and in the being corrupted boundless something continuous indeed, being,

καὶ οὐκ ἔστιν οὔτε γί γνεσθαι μὴ γεγονός τι οὔτε

and not is neither to become to be born not having happened something nor

γεγονέναι μὴ γιγνόμενόν τι, ὁμοίως δὲ καὶ ἐπὶ τοῦ φθείρεσθαι

having become not becoming something, similarly and also upon the being corrupted

καὶ ἐπὶ τοῦ ἐφθάρθαι· αἰεὶ γὰρ ἔσται τοῦ μὲν φθείρεσθαι τὸ

and upon the being decayed. always for it will be of the indeed to be destroyed the

ἐφθάρθαι πρότερον, τοῦ δ' ἐφθάρθαι τὸ φθείρεσθαι.

having been destroyed before, of the but to be destroyed the to be destroyed.

φανερὸν οὖν ὅτι καὶ τὸ γεγονὸς ἀνάγκη γίγνεσθαι

evident therefore that and the having happened necessity to come into being

πρότερον καὶ τὸ γιγνόμενον γεγονέναι· πᾶν γὰρ μέγεθος καὶ

before and the becoming having happened; all for magnitude and

πᾶς χρόνος ἀεὶ διαιρετά.

every time always divisible.

ὥστ' ἐν ᾧ ἂν ᾖ, οὐκ ἂν εἴη ὡς πρώτῳ.

so that in which may be, not would be as first.

Κεφάλαιο 7

Chapter 7

Ἐπεὶ δὲ πᾶν τὸ κινούμενον ἐν χρόνῳ κινεῖται, καὶ ἐν τῷ πλείονι

Since but all the moving in time is moved, and in the greater

μεῖζον μέγεθος, ἐν τῷ ἀπείρῳ χρόνῳ ἀδύνατόν ἐστιν πεπερασμένην

larger size, in the infinite time it is impossible to be finite

κινεῖσθαι, μὴ τὴν αὐτὴν αἰεὶ καὶ τῶν ἐκείνης τι κινούμενον,

to be moved, not the same always and of that something moving,

ἀλλ' ἐν ἅπαντι ἅπασαν. ὅτι μὲν οὖν εἴ τι ἰσοταχῶς

but in all entirety. That on the one hand, therefore if something uniformly

κινοῖτο, ἀνάγκη τὸ πεπερασμένον ἐν πεπερασμένῳ κινεῖσθαι,

moves, it is necessary the finite in finite to be moved,

δῆλον (ληφθέντος γὰρ μορίου ὃ καταμετρήσει τὴν ὅλην,

it is clear (since for part which will measure the whole,

ἐν ἴσοις χρόνοις τοσούτοις ὅσα τὰ μόριά ἐστιν, τὴν ὅλην

in equal times such as many the parts there are, the whole

κεκίνηται, ὥστ' ἐπεὶ ταῦτα πεπέρανται καὶ τῷ πόσον ἕκαστον

is moved, so that since these are all finite and the quantity each

καὶ τῷ ποσάκις ἅπαντα, καὶ ὁ χρόνος ἂν εἴη πεπερασμένος·

and the how many times all, and the time if it were finite.

τοσαυτάκις γὰρ ἔσται τοσοῦτος, ὅσος ὁ τοῦ μορίου χρόνος

as many times for it will be so much, as much as the of the part time

πολλαπλασιασθεὶς τῷ πλήθει τῶν μορίων)·

multiplied by the number of the parts).

ἀλλὰ δὴ κἂν μὴ ἰσοταχῶς, διαφέρει οὐθέν. ἔστω γὰρ ἐφ' ἧς

but indeed even if not uniformly, it differs nothing. let it be for on which

τὸ AB διάστημα πεπερασμένον, ὃ κεκίνηται [238a] ἐν τῷ ἀπείρῳ,

the AB distance finite, which moves [238a] in the infinite,

καὶ ὁ χρόνος ἄπειρος ἐφ' οὗ τὸ ΓΔ. εἰ δὴ ἀνάγκη πρότερον

and the time is infinite in which the GD. if indeed necessity earlier

ἕτερον ἑτέρου κεκινῆσθαι (τοῦτο δὲ δῆλον, ὅτι τοῦ χρόνου ἐν

another of another to move (this but is clear, that of the time in

τῷ προτέρῳ καὶ ὑστέρῳ ἕτερον κεκίνηται· ἀεὶ γὰρ ἐν τῷ πλείονι

the earlier and latter another is moved; always for in the greater

ἕτερον ἔσται κεκινημένον, ἐάν τε ἰσοταχῶς ἐάν τε μὴ ἰσοταχῶς

another will be in motion, if and uniformly if and not uniformly

μεταβάλλῃ, καὶ ἐάν τε ἐπιτείνῃ ἡ κίνησις ἐάν τε ἀνιῇ ἐάν τε

changes, and if and intensifies the motion if and rises if and

μένῃ, οὐθὲν ἧττον), εἰλήφθω δή τι τοῦ AB διαστήματος, τὸ

stays, nothing less), let it be taken indeed what of the AB interval, the

AE, ὃ καταμετρήσει τὴν AB. τοῦτο δὴ τοῦ ἀπείρου ἔν τινι

AE, which will measure the AB. this indeed of the infinite in some

ἐγένετο χρόνῳ· ἐν ἀπείρῳ γὰρ οὐχ οἷόν τε· τὸ γὰρ ἅπαν ἐν

came to be time; in the infinite for not possible at all; the for all in

ἀπείρῳ. καὶ πάλιν ἕτερον δὴ ἐὰν λάβω ὅσον τὸ AE, ἀνάγκη

the infinite. And again another indeed if I take as much as the AE, necessity

ἐν πεπερασμένῳ χρόνῳ· τὸ γὰρ ἅπαν ἐν ἀπείρῳ. καὶ οὕτω δὴ

in finite time; the for all in the infinite. And thus indeed

λαμβάνων, ἐπειδὴ τοῦ μὲν ἀπείρου οὐθὲν ἔστι μόριον ὃ

taking, since of the on one hand infinite nothing is a part which

καταμετρήσει (ἀδύνατον γὰρ τὸ ἄπειρον εἶναι ἐκ πεπερασμένων καὶ

will measure (impossible for the infinite to be from finite and

ἴσων καὶ ἀνίσων, διὰ τὸ καταμετρηθήσεσθαι τὰ πεπερασμένα

equal and unequal, because of the to be measured the finite

πλήθει καὶ μεγέθει ὑπό τινος ἑνός, ἐάν τε ἴσα ᾖ ἐάν τε

in quantity and in size under of some one, if and equal may be if and

ἄνισα, ὡρισμένα δὲ τῷ μεγέθει, οὐθὲν ἧττον), τὸ δὲ διάστημα τὸ

unequal, certain but to the size, nothing less), the but interval the

πεπερασμένον ποσοῖς τοῖς AE μετρεῖται, ἐν πεπερασμένῳ ἂν χρόνῳ

finite quantities the AE is measured, in finite if time

τὸ AB κινοῖτο (ὡσαύτως δὲ καὶ ἐπὶ ἠρεμήσεως)· ὥστε οὔτε

the AB would move (similarly but also during rest)· so that neither

γίγνεσθαι οὔτε φθείρεσθαι οἷόν τε ἀεί τι τὸ αὐτὸ καὶ ἕν.

to become nor to be destroyed possible and always something the same and one.

ὁ αὐτὸς δὲ λόγος καὶ ὅτι οὐδ' ἐν πεπερασμένῳ χρόνῳ ἄπειρον

the same but argument and that not even in finite time infinite

οἷόν τε κινεῖσθαι οὐδ' ἠρεμίζεσθαι, οὔθ' ὁμαλῶς κινούμενον οὔτ'

possible and to be moved nor to be at rest, nor smoothly moving nor

ἀνωμάλως. ληφθέντος γάρ τινος μέρους ὃ ἀναμετρήσει τὸν

unevenly. taken for something part which measures the

ὅλον χρόνον, ἐν τούτῳ ποσόν τι διέξεισιν τοῦ μεγέθους καὶ

whole time, in to this amount some it passes through of the size and

οὐχ ὅλον (ἐν γὰρ τῷ παντὶ τὸ ὅλον), καὶ πάλιν ἐν τῷ ἴσῳ ἄλλο,

not whole (in for the all the whole), and again in the equal other,

καὶ ἐν ἑκάστῳ ὁμοίως, εἴτε ἴσον εἴτε ἄνισον τῷ ἐξ ἀρχῆς·

and in each likewise, whether equal or unequal to that out of beginning;

διαφέρει γὰρ οὐδέν, εἰ μόνον πεπερασμένον ἕκαστον· δῆλον γὰρ

it differs for nothing, if only finite each; evident for

ὡς ἀναιρουμένου τοῦ χρόνου τὸ ἄπειρον οὐκ ἀναιρεθήσεται,

as being taken away of the time the infinite not will be eliminated,

πεπερασμένης τῆς ἀφαιρέσεως γιγνομένης καὶ τῷ ποσῷ καὶ τῷ

of the finite the subtraction becoming and the amount and the

ποσάκις· ὥστ' οὐ δίεισιν ἐν πεπερασμένῳ χρόνῳ τὸ ἄπειρον.

numerous; so that not it passes through in finite time the infinite.

οὐδέν τε διαφέρει τὸ μέγεθος ἐπὶ θάτερα ἢ ἐπ' ἀμφότερα εἶναι

nothing and differs the magnitude in one side or on both sides to be

ἄπειρον· ὁ γὰρ αὐτὸς ἔσται λόγος.

infinite; the for he will be word.

ἀποδεδειγμένων δὲ τούτων φανερὸν ὅτι οὐδὲ τὸ πεπερασμένον

having been demonstrated but of these evident that nor the finite

μέγεθος τὸ ἄπειρον ἐνδέχεται διελθεῖν ἐν πεπερασμένῳ διὰ
magnitude the infinite is possible to pass through in the finite through
τὴν αὐτὴν αἰτίαν· ἐν γὰρ τῷ μορίῳ τοῦ χρόνου πεπερασμένον
the same cause; in for the part of the time finite
δίεισι, καὶ ἐν ἑκάστῳ ὡσαύτως, ὥστ' ἐν τῷ παντὶ πεπερασμένον.
it passes, and in each likewise, so that in the whole finite.

ἐπεὶ δὲ τὸ πεπερασμένον οὐ δίεισι τὸ ἄπειρον [238b] ἐν πεπερασμένῳ
since but the finite not goes through the infinite [238b] in finite
χρόνῳ, δῆλον ὡς οὐδὲ τὸ ἄπειρον τὸ πεπερασμένον· εἰ γὰρ τὸ
time, it is clear that neither the infinite the finite; if indeed the
ἄπειρον τὸ πεπερασμένον, ἀνάγκη καὶ τὸ πεπερασμένον διιέναι
infinite the finite, necessity and the finite to pass through
τὸ ἄπειρον. οὐδὲν γὰρ διαφέρει ὁποτερονοῦν εἶναι τὸ κινούμενον·
the infinite. nothing indeed differs in any way to be the moving;
ἀμφοτέρως γὰρ τὸ πεπερασμένον δίεισι τὸ ἄπειρον. ὅταν γὰρ
either way indeed the finite goes through the infinite. when indeed
κινῆται τὸ ἄπειρον ἐφ' ᾧ τὸ Α, ἔσται τι αὐτοῦ κατὰ τὸ
it moves the infinite over which the A, it will be some of it according to the
Β τὸ πεπερασμένον, οἷον τὸ ΓΔ, καὶ πάλιν ἄλλο καὶ ἄλλο, καὶ
B, the finite, such as the GD, and again another and another, and
αἰεὶ οὕτως. ὥσθ' ἅμα συμβήσεται τὸ ἄπειρον κεκινῆσθαι
always thus. so that together it will happen the infinite to be set in motion
τὸ πεπερασμένον καὶ τὸ πεπερασμένον διεληλυθέναι τὸ ἄπειρον·
the finite and the finite having passed through the infinite.
οὐδὲ γὰρ ἴσως δυνατὸν ἄλλως τὸ ἄπειρον κινηθῆναι τὸ πεπερασμένον
nor for perhaps possible otherwise the infinite to be moved the finite

ἢ τῷ τὸ πεπερασμένον διιέναι τὸ ἄπειρον, ἢ φερόμενον ἢ

or by the finite to pass through the infinite, or carried or

ἀναμετροῦν. ὥστ' ἐπεὶ τοῦτ' ἀδύνατον, οὐκ ἂν διίοι τὸ

measured. Thus since this is impossible, not if it would pass through the

ἄπειρον τὸ πεπερασμένον. ἀλλὰ μὴν οὐδὲ τὸ ἄπειρον ἐν πεπερασμένῳ

infinite the finite. But indeed nor the infinite in finite

χρόνῳ τὸ ἄπειρον· δίεισιν· εἰ γὰρ τὸ ἄπειρον, καὶ τὸ

time the infinite. It passes through. If for the infinite, and the

πεπερασμένον· ἐνυπάρχει γὰρ τῷ ἀπείρῳ τὸ πεπερασμένον.

finite; exists for in the infinite the finite.

ἔτι δὲ καὶ τοῦ χρόνου ληφθέντος ἡ αὐτὴ ἔσται ἀπόδειξις.

Additionally and also the time taken the same will be proof.

ἐπεὶ δ' οὔτε τὸ πεπερασμένον τὸ ἄπειρον οὔτε τὸ ἄπειρον τὸ

Since but neither the finite the infinite neither the infinite the

πεπερασμένον οὔτε τὸ ἄπειρον τὸ ἄπειρον ἐν πεπερασμένῳ χρόνῳ

finite nor the infinite the infinite in finite time

κινεῖται, φανερὸν ὅτι οὐδὲ κίνησις ἔσται ἄπειρος ἐν πεπερασμένῳ

moves, evident that neither motion will be infinite in finite

χρόνῳ· τί γὰρ διαφέρει τὴν κίνησιν ἢ τὸ μέγεθος ποιεῖν ἄπειρον;

time. what for differs the motion or the magnitude to make infinite?

ἀνάγκη γάρ, εἰ ὁποτερονοῦν, καὶ θάτερον εἶναι ἄπειρον· πᾶσα γὰρ

necessity for, if either, and the other to be infinite. every for

φορὰ ἐν τόπῳ.

change in place.

Κεφάλαιο 8

Chapter 8

Ἐπεὶ δὲ πᾶν ἢ κινεῖται ἢ ἠρεμεῖ τὸ πεφυκὸς ὅτε πέφυκε καὶ

Since but everything either moves or rests the natural when has become and

οὗ καὶ ὡς, ἀνάγκη τὸ ἱστάμενον ὅτε ἵσταται κινεῖσθαι· εἰ γὰρ

of which and as, necessity the standing when stands to be moved. If for

μὴ κινεῖται, ἠρεμήσει, ἀλλ' οὐκ ἐνδέχεται ἠρεμίζεσθαι τὸ ἠρεμοῦν.

not moves, it will rest, but not it is possible to be at rest the resting.

τούτου δ' ἀποδεδειγμένου φανερὸν ὅτι καὶ ἐν χρόνῳ ἵστασθαι

Of this but having been proven it is clear that also in time to stand

ἀνάγκη (τὸ γὰρ κινούμενον ἐν χρόνῳ κινεῖται, τὸ δ' ἱστάμενον

is necessary (the for moving in time moves, the but standing

δέδεικται κινούμενον, ὥστε ἀνάγκη ἐν χρόνῳ ἵστασθαι)·

has been demonstrated moving, so that necessity in time to stand).

ἔτι δ' εἰ τὸ μὲν θᾶττον καὶ βραδύτερον ἐν χρόνῳ λέγομεν,

still but if the on the one hand faster and slower in time we say,

ἵστασθαι δ' ἔστιν θᾶττον καὶ βραδύτερον. ἐν ᾧ δὲ χρόνῳ

to stand but it is faster and slower. in which but time

πρώτῳ τὸ ἱστάμενον ἵσταται, ἐν ὁτῳοῦν ἀνάγκη τούτου ἵστασθαι.

first the standing stands, in any necessity of this to stand.

διαιρεθέντος γὰρ τοῦ χρόνου εἰ μὲν ἐν μηδετέρῳ τῶν

having been divided for of the time if on the one hand in neither of the

μερῶν ἵσταται, οὐδ' ἐν τῷ ὅλῳ, ὥστ' οὐκ ἂν ἵσταιτο τὸ ἱστάμενον·

parts it stands, nor in the whole, so that it cannot be stand; the standing;

εἰ δ' ἐν θατέρῳ, οὐκ ἂν ἐν πρώτῳ τῷ ὅλῳ ἵσταιτο· καθ'

if but in another, it cannot be in first the whole it could stand; according to

ἕτερον γὰρ ἐν τούτῳ ἵσταται, καθάπερ ἐλέχθη καὶ ἐπὶ τοῦ

another for in this it stands, as it was said and upon the

κινουμένου πρότερον. ὥσπερ δὲ τὸ κινούμενον οὐκ ἔστιν [239a]

moving earlier. just as but the moving is not it is [239a]

ἐν ᾧ πρώτῳ κινεῖται, οὕτως οὐδ' ἐν ᾧ ἵσταται τὸ ἱστάμενον·

in which first it moves, thus neither in which it stands the standing one.

οὔτε γὰρ τοῦ κινεῖσθαι οὔτε τοῦ ἵστασθαί ἐστίν τι πρῶτον.

nor for of moving nor of standing is something first.

ἔστω γὰρ ἐν ᾧ πρώτῳ ἵσταται ἐφ' ᾧ τὸ AB. τοῦτο δὴ

let it be for in which first it stands upon which the AB. this indeed

ἀμερὲς μὲν οὐκ ἐνδέχεται εἶναι (κίνησις γὰρ οὐκ ἔστιν

indivisible on the one hand not is possible to be (motion for not is

ἐν τῷ ἀμερεῖ διὰ τὸ κεκινῆσθαί τι ἂν αὐτοῦ, τὸ δ'

in the indivisible because of the having moved something at all of it, the but

ἱστάμενον δέδεικται κινούμε νον)· ἀλλὰ μὴν εἰ διαιρετόν

standing has been shown moving one). but indeed if it is divisible,

ἐστιν, ἐν ὁτῳοῦν αὐτοῦ τῶν μερῶν ἵσταται· τοῦτο γὰρ δέδεικται

it is, in whatever of it of the parts it stands. this for has been shown

πρότερον, ὅτι ἐν ᾧ πρώτῳ ἵσταται, ἐν ὁτῳοῦν τῶν ἐκείνου

previously, that in which first it stands, in whatever of those that

ἵσταται. ἐπεὶ οὖν χρόνος ἐστὶν ἐν ᾧ πρώτῳ ἵσταται, καὶ οὐκ

stands. since therefore time is in which first stands, and not

ἄτομον, ἅπας δὲ χρόνος εἰς ἄπειρα μεριστός, οὐκ ἔσται ἐν ᾧ

indivisible, all but time into infinite divisible, not will be in which

πρώτῳ ἵσταται.

first stands.

οὐδὲ δὴ τὸ ἠρεμοῦν ὅτε πρῶτον ἠρέμησεν ἔστιν. ἐν ἀμερεῖ

and not indeed the being at rest when first it rested it is. in unmeasured

μὲν γὰρ οὐκ ἠρέμησεν διὰ τὸ μὴ εἶναι κίνησιν ἐν ἀτόμῳ,

on one hand for not it rested because of the not to be motion in an atom,

ἐν ᾧ δὲ τὸ ἠρεμεῖν, καὶ τὸ κινεῖσθαι (τότε γὰρ ἔφαμεν

in which but the being at rest, and the being moved (then for we said

ἠρεμεῖν, ὅτε καὶ ἐν ᾧ πεφυκὸς κινεῖσθαι μὴ κινεῖται τὸ

to be at rest, when and in which it is natural to be moved not to be moved the

πεφυκός)· ἔτι δὲ καὶ τότε λέγομεν ἠρεμεῖν, ὅταν ὁμοίως

nature). still but also then we say to be at rest, whenever similarly

ἔχῃ νῦν καὶ πρότερον, ὡς οὐχ ἑνί τινι κρίνοντες ἀλλὰ δυοῖν

it holds now and previously, as not one certain judging but of two

τοῖν ἐλαχίστοιν· ὥστ' οὐκ ἔσται ἐν ᾧ ἠρεμεῖ ἀμερές. εἰ

the smallest. so that not it will be in which it rests unmeasured. if

δὲ μεριστόν, χρόνος ἂν εἴη, καὶ ἐν ὁτῳοῦν αὐτοῦ τῶν μερῶν

but divided, time would be, and in anything of it of the parts

ἠρεμήσει. τὸν αὐτὸν γὰρ τρόπον δειχθήσεται ὃν καὶ ἐπὶ τῶν

will be at rest. the same for manner will be shown which also upon the

πρότερον·

previously.

ὥστ' οὐθὲν ἔσται πρῶτον. τούτου δ' αἴτιον ὅτι ἠρεμεῖ μὲν

so that nothing will be first. of this but the cause that is at rest indeed

καὶ κινεῖται πᾶν ἐν χρόνῳ, χρόνος δ' οὐκ ἔστι πρῶτος οὐδὲ

and moves everything in time, time but not is first nor

μέγεθος οὐδ' ὅλως συνεχὲς οὐδέν· ἅπαν γὰρ εἰς ἄπειρα μεριστόν.

greatness nor at all continuous nothing; all for into infinite divisible.

ἐπεὶ δὲ πᾶν τὸ κινούμενον ἐν χρόνῳ κινεῖται καὶ ἔκ τινος εἴς

since but all the moving in time is moved and from something into

τι μεταβάλλει, ἐν ᾧ χρόνῳ κινεῖται καθ' αὐτὸν καὶ

something changes, in which time is moved according to itself and

μὴ τῷ ἐν ἐκείνου τινί, ἀδύνατον τότε κατά τι εἶναι

not to the in that something, impossible then according to something to be

πρῶτον τὸ κινού μενον. τὸ γὰρ ἠρεμεῖν ἐστιν τὸ ἐν τῷ αὐτῷ εἶναι

first the moving thing. the for rest is the in the same to be

χρόνον τινὰ καὶ αὐτὸ καὶ τῶν μερῶν ἕκαστον. οὕτως γὰρ λέγομεν

time some and itself and of the parts each. thus for we say

ἠρεμεῖν, ὅταν ἐν ἄλλῳ καὶ ἄλλῳ τῶν νῦν ἀληθὲς ᾖ εἰπεῖν ὅτι

to rest, when in another and another of the present true may be to say that

ἐν τῷ αὐτῷ καὶ αὐτὸ καὶ τὰ μέρη. εἰ δὲ τοῦτ' ἔστι τὸ ἠρεμεῖν, οὐκ

in the same and itself and the parts. if but this is the rest, not

ἐνδέχεται τὸ μεταβάλλον κατά τι εἶναι ὅλον κατὰ τὸν

it is possible the changing according to something to be whole according to the

πρῶτον χρόνον· ὁ γὰρ χρόνος διαιρετὸς ἅπας, ὥστε ἐν ἄλλῳ καὶ

first time; the for time divisible all, so that in another and

ἄλλῳ αὐτοῦ μέρει ἀληθὲς ἔσται εἰπεῖν ὅτι ἐν ταὐτῷ ἐστιν καὶ αὐτὸ

another of it part true will be to say that in the same is and it

καὶ τὰ μέρη. εἰ γὰρ μὴ οὕτως ἀλλ' ἐν ἑνὶ μόνῳ τῶν νῦν, οὐκ ἔσται

and the parts. if for not thus but in one alone of the now, not will be

χρόνον οὐδένα κατά τι, ἀλλὰ κατὰ τὸ πέρας τοῦ

time none according to something, but according to the end of the

χρόνου. ἐν δὲ τῷ νῦν ἔστιν μὲν ἀεὶ κατά τι μὲν [239b]

time. in but the now it is indeed always according to something indeed [239b]

ὄν, οὐ μέντοι ἠρεμεῖ· οὔτε γὰρ κινεῖσθαι οὔτ' ἠρεμεῖν ἔστιν

being, not however it is at rest; nor for to be moved nor to be at rest is

ἐν τῷ νῦν, ἀλλὰ μὴ κινεῖσθαι μὲν ἀληθὲς ἐν τῷ νῦν καὶ

in to the now, but not to be moved on the one hand true in the now and

εἶναι κατά τι, ἐν χρόνῳ δ' οὐκ ἐνδέχεται εἶναι κατά

to be according to something, in time and not is possible to be according to

τι ἠρεμοῦν· συμβαίνει γὰρ τὸ φερόμενον ἠρεμεῖν.

something at rest. it happens for the being carried to be at rest.

Κεφάλαιο 9

Chapter 9

Ζήνων δὲ παραλογίζεται· εἰ γὰρ αἰεί, φησίν, ἠρεμεῖ πᾶν [ἢ κινεῖται]

Zeno but reasons absurdly; if for always, he says, it rests all or it moves]

ὅταν ᾖ κατὰ τὸ ἴσον, ἔστιν δ' αἰεὶ τὸ φερόμενον ἐν τῷ

when it is according to the equal, there is but always the movable in the

νῦν, ἀκίνητον τὴν φερομένην εἶναι ὀϊστόν. τοῦτο δ' ἐστὶ ψεῦδος·

now, unmoving the being moved to be an arrow. This but is a falsehood;

οὐ γὰρ σύγκειται ὁ χρόνος ἐκ τῶν νῦν τῶν ἀδιαιρέτων, ὥσπερ

not for it is composed the time from the now and the indivisible, just as

οὐδ' ἄλλο μέγεθος οὐδέν.

neither any other magnitude nothing.

τέτταρες δ' εἰσὶν οἱ λόγοι περὶ κινήσεως Ζήνωνος οἱ

Four but there are the arguments concerning motion of Zeno the

παρέχοντες τὰς δυσκολίας τοῖς λύουσιν, πρῶτος μὲν ὁ περὶ

ones presenting the difficulties to those who solve, the first indeed the concerning

τοῦ μὴ κινεῖσθαι διὰ τὸ πρότερον εἰς τὸ ἥμισυ δεῖν ἀφικέσθαι

the not moving due to the prior to the half having to arrive

τὸ φερόμενον ἢ πρὸς τὸ τέλος, περὶ οὗ διείλομεν ἐν τοῖς

the movable or towards the end, concerning which we have divided in the

πρότερον λόγοις.

earlier discourses.

δεύτερος δ' ὁ καλούμενος Ἀχιλλεύς· ἔστι δ' οὗτος, ὅτι τὸ

second that the called Achilles. is that this, because the

βραδύτατον οὐδέποτε καταληφθήσεται θέον ὑπὸ τοῦ ταχίστου·

slowest never will be caught running by the fastest.

ἔμπροσθεν γὰρ ἀναγκαῖον ἐλθεῖν τὸ διῶκον ὅθεν ὥρμησεν τὸ

Before for it is necessary to come the pursuing from where it has started the

φεῦγον, ὥστε ἀεί τι προέχειν ἀναγκαῖον τὸ βραδύτερον.

fleeing, so that always something takes precedence it is necessary the slower.

ἔστιν δὲ καὶ οὗτος ὁ αὐτὸς λόγος τῷ διχοτομεῖν, διαφέρει

It is and also this the same reasoning in doing the bisection, differs

δ' ἐν τῷ διαιρεῖν μὴ δίχα τὸ προσλαμβανόμενον μέγεθος. τὸ μὲν

and in the dividing not apart the taken size. The one

οὖν μὴ καταλαμβάνε σθαι τὸ βραδύτερον συμβέβηκεν ἐκ τοῦ

therefore not occurs to arise the slower has happened from the

λόγου, γίγνεται δὲ παρὰ ταὐτὸ τῇ διχοτομίᾳ (ἐν ἀμφοτέροις

reasoning, it becomes and alongside the same to the bisection (in both

γὰρ συμβαίνει μὴ ἀφικνεῖσθαι πρὸς τὸ πέρας διαιρουμένου πως

for it happens not to arrive at the end being divided how

τοῦ μεγέθους· ἀλλὰ πρόσκειται ἐν τούτῳ ὅτι οὐδὲ τὸ τάχιστον

of the size; but it pertains in this that neither the fastest

τετραγῳδημένον ἐν τῷ διώκειν τὸ βραδύτατον), ὥστ' ἀνάγκη καὶ

squared in the pursuing the slowest), so that necessity and

τὴν λύσιν εἶναι τὴν αὐτήν. τὸ δ' ἀξιοῦν ὅτι τὸ προέχον οὐ

the solution to be the same. the but it is necessary that the prevailing not

καταλαμβάνεται, ψεῦδος· ὅτε γὰρ προέχει, οὐ καταλαμβάνεται·

is understood, a falsehood; since for it prevails, not is understood;

ἀλλ' ὅμως καταλαμβάνεται, εἴπερ δώσει διεξιέναι τὴν

but nevertheless it is understood, if it would allow to proceed the

πεπερασμένην. οὗτοι μὲν οὖν οἱ δύο λόγοι,

finite. these indeed thus the two arguments,

τρίτος δ' ὁ νῦν ῥηθείς, ὅτι ἡ ὀϊστὸς φερομένη ἕστηκεν.

a third but the now mentioned, that the arrow being carried has stood still.

συμβαίνει δὲ παρὰ τὸ λαμβάνειν τὸν χρόνον συγκεῖσθαι ἐκ

it happens but in addition to the taking the time to be composed of

τῶν νῦν· μὴ διδομένου γὰρ τούτου οὐκ ἔσται ὁ συλλογισμός.

the now; not allowing for this there will not be the reasoning

τέταρτος δ' ὁ περὶ τῶν ἐν τῷ σταδίῳ κινουμένων ἐξ ἐναντίας

fourth but the concerning the in the stadium moving from opposite

ἴσων ὄγκων παρ' ἴσους, τῶν μὲν ἀπὸ τέλους τοῦ σταδίου

equal masses from equal, the on the one hand from the end of the stadium

τῶν δ' ἀπὸ μέσου, ἴσῳ τάχει, ἐν ᾧ συμβαίνειν [240a] οἴεται

the but from the middle, equal speed, in which happens [240a] he/she thinks

ἴσον εἶναι χρόνον τῷ διπλασίῳ τὸν ἥμισυν. ἔστι δ' ὁ παραλογισμὸς

equal to be time the double the half. it is but the paradox

ἐν τῷ τὸ μὲν παρὰ κινούμενον τὸ δὲ παρ'

in the the on the one hand from moving the on the other hand from

ἠρεμοῦν τὸ ἴσον μέγεθος ἀξιοῦν τῷ ἴσῳ τάχει τὸν ἴσον φέρεσθαι

resting the equal magnitude demands the equal speed the equal to carry

χρόνον· τοῦτο δ' ἐστὶ ψεῦδος. οἷον ἔστωσαν οἱ ἑστῶτες ἴσοι

time. this but is a falsehood. such as let them be the standing equal

ὄγκοι ἐφ' ὧν τὰ AA, οἱ δ' ἐφ' ὧν τὰ BB ἀρχόμενοι ἀπὸ τοῦ

masses upon which the AA, the but upon of whom the BB beginning from the

μέσου, ἴσοι τὸν ἀριθμὸν τούτοις ὄντες καὶ τὸ μέγεθος, οἱ δ'

middle, are equal the number to these being and the size, the but

ἐφ' ὧν τὰ ΓΓ ἀπὸ τοῦ ἐσχάτου, ἴσοι τὸν ἀριθμὸν ὄντες τούτοις

on which the GG from the last, are equal the number being to these

καὶ τὸ μέγεθος, καὶ ἰσοταχεῖς τοῖς B. συμβαίνει δὴ τὸ πρῶτον

and the size, and equidistant to the B. it happens indeed the first

B ἅμα ἐπὶ τῷ ἐσχάτῳ εἶναι καὶ τὸ πρῶτον Γ, παρ' ἄλληλα

B together at the last to be and the first G, among one another

κινουμένων.

moving.

συμβαίνει δὲ τὸ Γ παρὰ πάντα [τὰ B] διεξεληλυθέναι, τὸ δὲ B

happens but the G from all the B] to have been completed, the but B

παρὰ τὰ ἡμίση· ὥστε ἥμισυν εἶναι τὸν χρόνον· ἴσον γὰρ ἑκάτερόν

from the half· so that half to be the time· equal for each

ἐστιν παρ' ἕκαστον. ἅμα δὲ συμβαίνει τὸ πρῶτον B παρὰ

is from each. at the same time but happens the first B from

πάντα τὰ Γ παρεληλυθέναι· ἅμα γὰρ ἔσται τὸ πρῶτον

all the G to have been completed· at the same time for there will be the first

Γ καὶ τὸ πρῶτον B ἐπὶ τοῖς ἐναντίοις ἐσχάτοις, [ἴσον χρόνον παρ'

G and the first B upon the opposite last, equal time from

ἕκαστον γιγνόμενον τῶν B ὅσον περ τῶν A, ὥς φησιν,]

each occurring of the B as far as concerning the A, as he says,]

διὰ τὸ ἀμφότερα ἴσον χρόνον παρὰ τὰ A γίγνεσθαι. ὁ μὲν

through the both equal time from the A to become. the indeed

οὖν λόγος οὗτός ἐστιν, συμβαίνει δὲ παρὰ τὸ εἰρημένον ψεῦδος.

therefore argument this is, happens but from the said falsehood.

οὐδὲ δὴ κατὰ τὴν ἐν τῇ ἀντιφάσει μεταβολὴν οὐθὲν ἡμῖν

neither indeed according to the in the contradiction change nothing to us

ἔσται ἀδύνατον, οἷον εἰ ἐκ τοῦ μὴ λευκοῦ εἰς τὸ λευκὸν μετα

will be impossible, such as if from the not white into the white after

βάλλει καὶ ἐν μηδετέρῳ ἐστίν, ὡς ἄρα οὔτε λευκὸν ἔσται

throws and in neither is, as therefore neither white will be

οὔτε οὐ λευκόν· οὐ γὰρ εἰ μὴ ὅλον ἐν ὁποτερῳοῦν ἐστιν, οὐ

nor not white. not for if not whole in either is, not

λεχθήσεται λευκὸν ἢ οὐ λευκόν· λευκὸν γὰρ λέγομεν ἢ οὐ

will be said white or not white. white for we say or not

λευκὸν οὐ τῷ ὅλον εἶναι τοιοῦτον, ἀλλὰ τῷ τὰ πλεῖστα ἢ

white not in the whole to be such, but in the the most or

τὰ κυριώτατα μέρη· οὐ ταὐτὸ δ' ἐστὶν μὴ εἶναί τε ἐν τούτῳ

the principal parts. not the same but it is not to be and in this

καὶ μὴ εἶναι ἐν τούτῳ ὅλον. ὁμοίως δὲ καὶ ἐπὶ τοῦ ὄντος καὶ ἐπὶ

and not to be in to this whole. similarly but and upon the being and upon

τοῦ μὴ ὄντος καὶ τῶν ἄλλων τῶν κατ' ἀντίφασιν· ἔσται

the not being and of the others of the according to contradiction. it will be

μὲν γὰρ ἐξ ἀνάγκης ἐν θατέρῳ τῶν ἀντικειμένων, ἐν

on the one hand for out of necessity in the other of the opposites, in

οὐδετέρῳ δ' ὅλον αἰεί.

neither but whole always.

πάλιν δ' ἐπὶ τοῦ κύκλου καὶ ἐπὶ τῆς σφαίρας καὶ ὅλως τῶν
again and upon the circle and upon the sphere and altogether of the
ἐν αὐτοῖς κινουμένων, ὅτι συμβήσεται αὐτὰ ἠρεμεῖν· ἐν γὰρ
in themselves moving, that it will happen them to be at rest; in for
τῷ αὐτῷ τόπῳ χρόνον τινὰ ἔσται καὶ αὐτὰ καὶ τὰ μέρη,
the same place time a certain it will be and them and the parts,
ὥστε ἠρεμήσει ἅμα καὶ κινήσεται. πρῶτον μὲν γὰρ τὰ
so that it will be at rest together and it will move. First indeed for the
μέρη οὐκ ἔστιν ἐν τῷ αὐτῷ οὐθένα χρό [240b] νον, εἶτα καὶ τὸ
parts are not there in the same anywhere time [240b] any, then and the
ὅλον μεταβάλλει αἰεὶ εἰς ἕτερον· οὐ γὰρ ἡ αὐτή ἐστιν ἡ ἀπὸ
whole it changes always into another; not for the same is the from
τοῦ A λαμβανομένη περιφέρεια καὶ ἡ ἀπὸ τοῦ B καὶ τοῦ Γ καὶ
the A taken circumference and the from the B and the C and
τῶν ἄλλων ἑκάστου σημείων, πλὴν ὡς ὁ μουσικὸς ἄνθρωπος καὶ
of the others of each point, except as the musician person and
ἄνθρωπος, ὅτι συμβέβηκεν. ὥστε μεταβάλλει αἰεὶ ἡ ἑτέρα εἰς τὴν
person, that has happened. so that it changes always the other into the
ἑτέραν, καὶ οὐδέποτε ἠρεμήσει. τὸν αὐτὸν δὲ τρόπον καὶ ἐπὶ τῆς
other, and never will be at rest. the same but manner and upon the
σφαίρας καὶ ἐπὶ τῶν ἄλλων τῶν ἐν αὐτοῖς κινουμένων.
sphere and upon the others of the in themselves moving.

Κεφάλαιο 10

Chapter 10

Ἀποδεδειγμένων δὲ τούτων λέγομεν ὅτι τὸ ἀμερὲς οὐκ ἐνδέχεται
Of the proven but these we say that the part not is possible

κινεῖσθαι πλὴν κατὰ συμβεβηκός, οἷον κινουμένου τοῦ
to be moved except according to the contingent, such as when moved of the
σώματος ἢ τοῦ μεγέθους τῷ ἐνυπάρχειν, καθάπερ ἂν εἰ τὸ
body or of the magnitude to exist within, just as if it is the
ἐν τῷ πλοίῳ κινοῖτο ὑπὸ τῆς τοῦ πλοίου φορᾶς ἢ τὸ μέρος τῇ
in the ship were moving by the of the ship load or the part to the
τοῦ ὅλου κινήσει. (ἀμερὲς δὲ λέγω τὸ κατὰ ποσὸν ἀδιαίρετον.)
of the whole motion. (The part but I say the according to quantity indivisible.)
καὶ γὰρ αἱ τῶν μερῶν κινήσεις ἕτεραί εἰσι κατ' αὐτά
And for the of the parts movements are other they are according to themselves
τε τὰ μέρη καὶ κατὰ τὴν τοῦ ὅλου κίνησιν. ἴδοι δ'
and the parts and according to the of the whole motion. Let him see but
ἄν τις ἐπὶ τῆς σφαίρας μάλιστα τὴν διαφοράν· οὐ γὰρ ταὐτὸν
if anyone on the sphere especially the difference; not for the same
τάχος ἐστὶ τῶν τε πρὸς τῷ κέντρῳ καὶ τῶν ἐκτὸς καὶ τῆς
speed is of the and towards the center and of the outside and of the
ὅλης, ὡς οὐ μιᾶς οὔσης κινήσεως. καθάπερ οὖν εἴπομεν, οὕτω
whole, as not one being of motion. just as therefore we said, thus
μὲν ἐνδέχεται κινεῖσθαι τὸ ἀμερὲς ὡς ὁ ἐν τῷ πλοίῳ
on one hand it is possible to be moved the indivisible as the in the boat
καθήμενος τοῦ πλοίου θέοντος, καθ' αὐτὸ δ' οὐκ ἐνδέχεται.
sitting of the boat, looking, according to itself but not it is possible.
μεταβαλλέτω γὰρ ἐκ τοῦ AB εἰς τὸ ΒΓ, εἴτ' ἐκ μεγέθους εἰς
let it change for from the AB to the BC, or from size to
μέγεθος εἴτ' ἐξ εἴδους εἰς εἶδος εἴτε κατ' ἀντίφασιν· ὁ
size or from form to form or according to contradiction: the
δὲ χρόνος ἔστω ἐν ᾧ πρώτῳ μεταβάλλει ἐφ' οὗ Δ. οὐκοῦν
but time let it be in which first it changes upon which D. therefore

ἀνάγκη αὐτὸ καθ' ὃν μεταβάλλει χρόνον ἢ ἐν τῷ AB εἶναι
necessity it according to which it changes time or in the AB to be
ἢ ἐν τῷ ΒΓ, ἢ τὸ μέν τι αὐτοῦ ἐν τούτῳ τὸ δ' ἐν
or in the BC, or the on one hand something of it in this the but in
θατέρῳ· πᾶν γὰρ τὸ μεταβάλλον οὕτως εἶχεν. ἐν ἑκατέρῳ μὲν
the other; all for the changing thus it had. in each on one side
οὖν οὐκ ἔσται τι αὐτοῦ· μεριστὸν γὰρ ἂν εἴη. ἀλλὰ
therefore not will be anything of it; a part for if it were. but
μὴν οὐδ' ἐν τῷ ΒΓ· μεταβεβληκὸς γὰρ ἔσται, ὑπόκειται
indeed not even in the BC; having changed for it will be, it is subject
δὲ μεταβάλλειν. λείπεται δὴ αὐτὸ ἐν τῷ AB εἶναι, καθ'
and to changing. it remains indeed itself in the AB to be, according to
ὃν μεταβάλλει χρόνον. ἠρεμήσει ἄρα· τὸ γὰρ ἐν τῷ αὐτῷ
which it changes time. it will rest therefore; the for in the same
εἶναι χρόνον τινὰ ἠρεμεῖν ἦν. ὥστ' οὐκ ἐνδέχεται τὸ ἀμερὲς
to be time some to be at rest it was. so not it is possible the partless
κινεῖσθαι οὐδ' ὅλως μεταβάλλειν· μοναχῶς γὰρ ἂν οὕτως ἦν
to be moved nor wholly to change· alone for it would thus was
αὐτοῦ κίνησις, εἰ ὁ χρόνος ἦν ἐκ τῶν νῦν· αἰεὶ γὰρ ἐν τῷ νῦν
of it motion, if the time was from the now· always for in the now
κεκινημένον ἂν ἦν καὶ μετα [241a] βεβληκός, ὥστε κινεῖσθαι
having moved it would be and after [241a] thrown, so as to be moved
μὲν μηδέποτε, κεκινῆσθαι δ' ἀεί. τοῦτο δ' ὅτι ἀδύνατον,
indeed never, to have moved but always. this but that it is impossible,
δέδεικται καὶ πρότερον· οὔτε γὰρ ὁ χρόνος ἐκ τῶν νῦν οὔθ'
it has been shown and previously· nor for the time from the now neither
ἡ γραμμὴ ἐκ στιγμῶν οὔθ' ἡ κίνησις ἐκ κινημάτων· οὐθὲν γὰρ
the line from moments nor the motion from movements· nothing for

ἄλλο ποιεῖ ὁ τοῦτο λέγων ἢ τὴν κίνησιν ἐξ ἀμερῶν, καθάπερ
else does the this saying or the motion from parts, just as
ἂν εἰ τὸν χρόνον ἐκ τῶν νῦν ἢ τὸ μῆκος ἐκ στιγμῶν.
it would if the time from the now or the length from moments.

ἔτι δὲ καὶ ἐκ τῶνδε φανερὸν ὅτι οὔτε στιγμὴν οὔτ' ἄλλο ἀδιαίρετον
yet but and from these evident that neither moment nor other indivisible
οὐθὲν ἐνδέχεται κινεῖσθαι. ἅπαν γὰρ τὸ κινούμενον ἀδύνατον
nothing is possible to be moved. all for the moving impossible
πρότερον μεῖζον κινηθῆναι αὑτοῦ, πρὶν ἢ ἴσον ἢ ἔλαττον. εἰ δὴ
prior greater to be moved itself, before or equal or less. if indeed
τοῦτο, φανερὸν ὅτι καὶ ἡ στιγμὴ ἔλαττον ἢ ἴσον κινηθήσεται
this, evident that also the moment less or equal will be moved
πρῶτον. ἐπεὶ δὲ ἀδιαίρετος, ἀδύνατον ἔλαττον κινηθῆναι πρότερον·
first. since but indivisible, impossible less to be moved prior·
ἴσην ἄρα αὐτῇ. ὥστε ἔσται ἡ γραμμὴ ἐκ στιγμῶν· αἰεὶ
equal therefore to itself. hence there will be the line from moments· always
γὰρ ἴσην κινουμένη τὴν πᾶσαν γραμμὴν στιγμὴ καταμετρήσει. εἰ δὲ
for equal moving the whole line moment will measure. if but
τοῦτο ἀδύνατον, καὶ τὸ κινεῖσθαι τὸ ἀδιαίρετον ἀδύνατον.
this is impossible, and the to be moved the indivisible is impossible.

ἔτι δ' εἰ ἅπαν ἐν χρόνῳ κινεῖται, ἐν δὲ τῷ νῦν μηθέν, ἅπας δὲ
Yet but if all in time is moving, in but the now nothing, all but
χρόνος διαιρετός, εἴη ἄν τις χρόνος ἐλάττων ὁτῳοῦν
time divisible, it would be perhaps someone time lesser of any
τῶν κινουμένων ἢ ἐν ᾧ κινεῖται ὅσον αὐτό. οὗτος μὲν γὰρ
of the moving or in which it moves as much as itself. This indeed for

ἔσται χρόνος ἐν ᾧ κινεῖται διὰ τὸ πᾶν ἐν χρόνῳ κινεῖσθαι,

will be time in which it moves through the whole in time to be moving,

χρόνος δὲ πᾶς διαιρετὸς δέδεικται πρότερον. εἰ δ' ἄρα στιγμὴ

time but all divisible has been shown beforehand. if but then instant

κινεῖται, ἔσται τις χρόνος ἐλάττων ἢ ἐν ᾧ αὐτὴν ἐκινήθη.

is moving, will be someone time lesser or in which itself was moved.

ἀλλὰ ἀδύνατον· ἐν γὰρ τῷ ἐλάττονι ἔλαττον ἀνάγκη κινεῖσθαι.

but impossible; in for the lesser lesser necessity to move.

ὥστε ἔσται διαιρετὸν τὸ ἀδιαίρετον εἰς τὸ ἔλαττον, ὥσπερ καὶ

so that it will be divisible the indivisible into the lesser, just as and

ὁ χρόνος εἰς τὸν χρόνον. μοναχῶς γὰρ ἂν κινοῖτο τὸ ἀμερὲς

the time into the time. alone for would move the indivisible

καὶ ἀδιαίρετον, εἰ ἦν ἐν τῷ νῦν κινεῖσθαι δυνατὸν τῷ ἀτόμῳ·

and undivided, if it were in the now to be moved possible for the atom;

τοῦ γὰρ αὐτοῦ λόγου ἐν τῷ νῦν κινεῖσθαι καὶ ἀδιαίρετόν τι

of the for its nature in the now to be moved and undivided something

κινεῖσθαι.

to be moved.

μεταβολὴ δ' οὐκ ἔστιν οὐδεμία ἄπειρος· ἅπασα γὰρ ἦν ἔκ τινος

change but not exists no infinite; all for was from some

εἴς τι, καὶ ἡ ἐν ἀντιφάσει καὶ ἡ ἐν ἐναντίοις. ὥστε τῶν

into something, and the in antithesis and the in opposites. so that of the

μὲν κατ' ἀντίφασιν ἡ φάσις καὶ ἡ ἀπόφασις

on one hand in accordance with antithesis the statement and the denial

πέρας (οἷον γενέσεως μὲν τὸ ὄν, φθορᾶς δὲ τὸ μὴ

limit (such as of generation on one hand the being, of corruption but the not

ὄν), τῶν δ' ἐν τοῖς ἐναντίοις τὰ ἐναντία· ταῦτα γὰρ ἄκρα
being), of the but in the opposites the opposites; these for extremes
τῆς μεταβολῆς, ὥστε καὶ ἀλλοιώσεως πάσης (ἐξ ἐναντίων γάρ
of the change, so that and of alteration all (from opposites for
τινων ἡ ἀλλοίωσις), ὁμοίως δὲ καὶ αὐξήσεως καὶ φθίσεως·
some the alteration), similarly but also of increase and decrease;
αὐξήσεως μὲν γὰρ τὸ πέρας τοῦ [241b] κατὰ τὴν οἰκείαν
of increase on one hand for the limit of [241b] according to the natural
φύσιν τελείου μεγέθους, φθίσεως δὲ ἡ τούτου ἔκστασις. ἡ δὲ
nature of the end size, of decay and the of this ecstasy. the and
φορὰ οὕτω μὲν οὐκ ἔσται πεπερασμένη· οὐ γὰρ πᾶσα
occurrence thus on the one hand not will be limited; not for every
ἐν ἐναντίοις· ἀλλ' ἐπειδὴ τὸ ἀδύνατον τμηθῆναι οὕτω, τῷ
in opposing things; but since the impossible to be divided in this way, by
μὴ ἐνδέχεσθαι τμηθῆναι (πλεοναχῶς γὰρ λέγεται τὸ ἀδύνατον),
not being possible to be divided (in excess for it is said the impossible),
οὐκ ἐνδέχεται τὸ οὕτως ἀδύνατον τέμνεσθαι, οὐδὲ ὅλως
not it is possible for the such impossible to be cut, nor at all
τὸ ἀδύνατον γενέσθαι γίγνεσθαι, οὐδὲ τὸ μεταβαλεῖν ἀδύνατον
the impossible to come to be to become, nor the to change impossible
ἐνδέχοιτ' ἂν μεταβάλλειν εἰς ὃ ἀδύνατον μεταβαλεῖν. εἰ οὖν
could if to change into what is impossible to change. if therefore
τὸ φερόμενον μεταβάλλοι εἴς τι, καὶ δυνατὸν ἔσται
the brought would change into something, and it is possible it will be
μεταβαλεῖν.
to change.

ὥστ' οὐκ ἄπειρος ἡ κίνησις, οὐδ' οἰσθήσεται τὴν ἄπειρον·

so that not infinite the motion, nor will be perceived the infinite.

ἀδύνα τον γὰρ διελθεῖν αὐτήν. ὅτι μὲν οὖν οὕτως οὐκ ἔστιν

impossible to for traverse it. that indeed therefore thus not is

ἄπειρος μεταβολὴ ὥστε μὴ ὡρίσθαι πέρασι, φανερόν. ἀλλ' εἰ

infinite change so that not to be defined limit, evident. but if

οὕτως ἐνδέχεται ὥστε τῷ χρόνῳ εἶναι ἄπειρον τὴν αὐτὴν

thus it is possible so that with time time to be infinite the same

οὖσαν καὶ μίαν, σκεπτέον. μὴ μιᾶς μὲν γὰρ γιγνομένης

being and one, it must be considered. not of one indeed for that is generated

οὐθὲν ἴσως κωλύει, οἷον εἰ μετὰ τὴν φορὰν ἀλλοίωσις εἴη

nothing perhaps prevents, such as if after the motion alteration there may be

καὶ μετὰ τὴν ἀλλοίωσιν αὔξησις καὶ πάλιν γένεσις· οὕτω γὰρ αἰεὶ

and after the alteration increase and again generation. thus for always

μὲν ἔσται τῷ χρόνῳ κίνησις, ἀλλ' οὐ μία διὰ τὸ μὴ

indeed there will be to time time motion, but not one because of the not

εἶναι μίαν ἐξ ἁπασῶν. ὥστε δὲ γίγνεσθαι μίαν, οὐκ ἐνδέχεται

to be one from all so that but to become one, not it is possible

ἄπειρον εἶναι τῷ χρόνῳ πλὴν μιᾶς· αὕτη δ' ἐστὶν ἡ κύκλῳ

infinite to be in the time except one. this and is the circular

φορά.

motion.